MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ADDER, DIGITAL, KEY CODE KBC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic adder module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:38Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.13 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KBC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/39-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

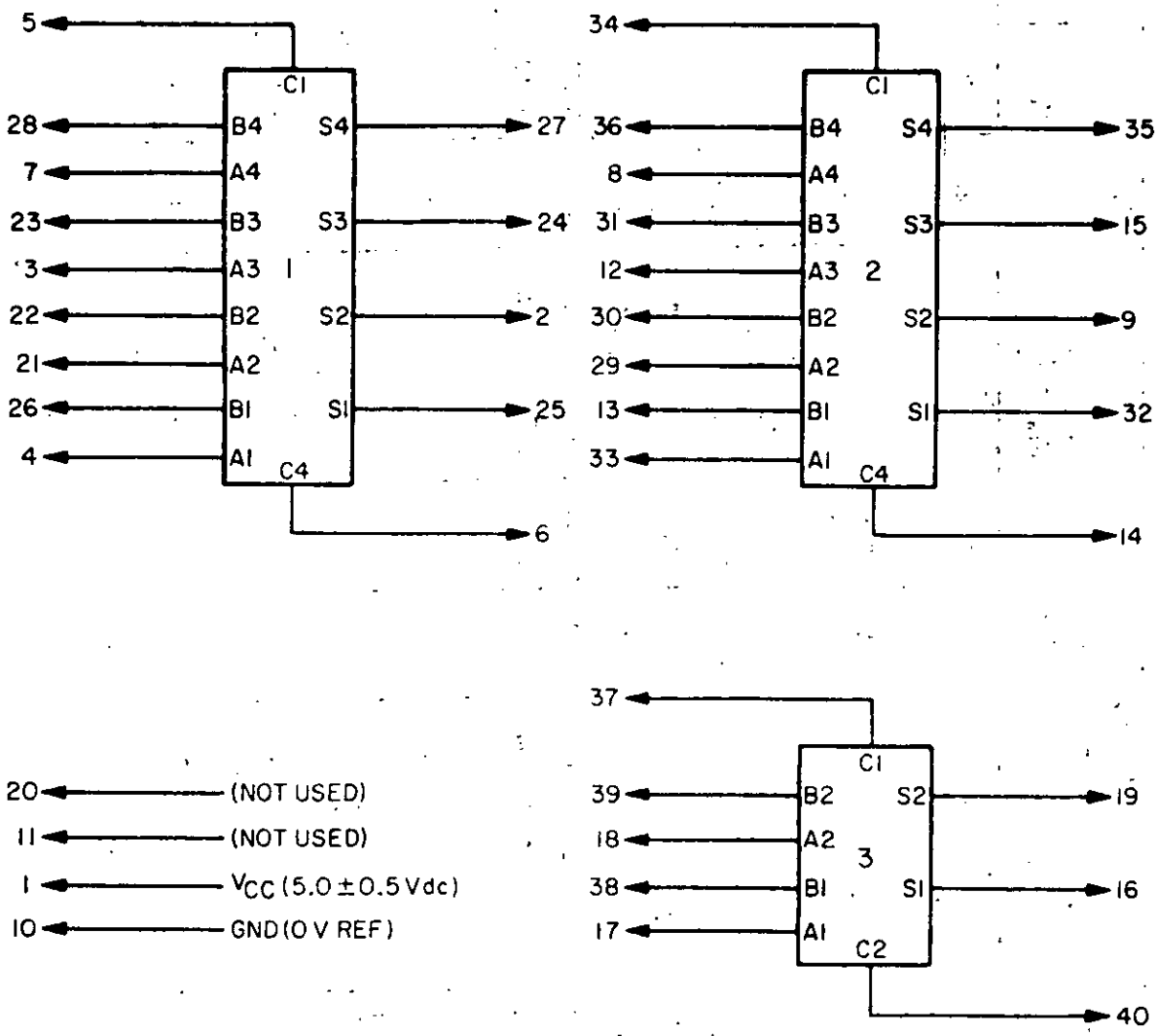
3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements specified in 3.10.1 through 3.10.3.

3.10.1 Test patterns. A set of unique input patterns shall be generated capable of detecting all nonredundant failures in the module logic network. The test patterns shall be generated by assuming "stuck at H state" and "stuck at L state" failures in each of the basic logic functions and applying input patterns to detect these failures. The unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the tests of tables I and XV satisfy or exceed these unique test patterns, no additional tests shall be added; otherwise the needed tests shall be added to table XV. "Stuck at H state" indicates a failure resulting in a steady high-level output. "Stuck at L state" indicates a failure resulting in a steady low-level output. Prior to any change in the module design which would necessitate a different set of test patterns to detect "stuck at H state" or "stuck at L state", the vendor shall be required to submit to the contracting activity and the Standard Electronic Modules Program Quality Assurance Activity the proposed new set of test patterns for approval.

Source: https://assist.dla.mil -- Downloaded: 20260410-20T15:38Z
Check the source to verify that this is the current version before use.



C1 = Carry input.
A1-A4, B1-B4 = Data inputs.
S1-S4 = Sum outputs.
C2, C4 = Carry outputs.

FIGURE 1. KBC module function diagram, adder.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:38Z
Check the source to verify that this is the current version before use.

* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A13	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	mV
B1,B6,B11	Input breakdown current	IV	S	0	100	0	100	0	100	0	100	µA
B2-B5, B7-B10, B12-B23	Input breakdown current	IV	S	0	200	0	200	0	200	0	200	µA
C1-C3	Isolation	V	S	10	∞	1	∞	10	∞	1	∞	MΩ
D1-D23	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E13	Output voltage, low level	VII	X	0	400	0	400	0	400	0	400	mV
F1-F13	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1,G6,G11	Input current, low level	VIII	X	-0.01	-0.44	-0.01	-0.44	-0.01	-0.44	-0.01	-0.44	mA
G2-G5, G7-G10, G12-G23	Input current, low level	VIII	X	-0.01	-0.88	-0.01	-0.88	-0.01	-0.88	-0.01	-0.88	mA
H1,H6,H11	Input current, high level	VIII	X	0	20	0	20	0	20	0	20	µA
H2-H5, H7-H10, H12-H23	Input current, high level	VIII	X	0	40	0	40	0	40	0	40	µA
J1-J13	Output short circuit current	VIII	X	-6	-120	-6	-120	-6	-120	-6	-120	mA
K1	Power supply current, low level	IX	S	10	117	10	117	10	117	10	117	mA
K2	Power supply current, high level	IX	S	10	102	10	102	10	102	10	102	mA

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L1-L13	Transition time (t_{TLH}), C_L = 50 pF	X	S	5	50	5	50	5	60	5	60	ns
L14-L26	Transition time (t_{TLH}), C_L = 150 pF	X	Q	10	110	10	110	10	120	10	120	ns
M1-M13	Transition time (t_{THL}), C_L = 50 pF	X	S	3	25	3	25	3	30	3	30	ns
M14-M26	Transition time (t_{THL}), C_L = 150 pF	X	Q	6	35	6	35	6	40	6	40	ns
N1-N35	Propagation delay time (t_{PLH}), C_L = 50 pF	X	S	5	28	5	28	5	30	5	30	ns
N36-N53	Propagation delay time (t_{PLH}), C_L = 50 pF	X	S	5	21	5	21	5	23	5	23	ns
N54-N88	Propagation delay time (t_{PLH}), C_L = 150 pF	X	Q	5	38	5	38	5	40	5	40	ns
N89-N106	Propagation delay time (t_{PLH}), C_L = 150 pF	X	Q	5	31	5	31	5	33	5	33	ns
P1-P35	Propagation delay time (t_{PHL}), C_L = 50 pF	X	S	5	28	5	28	5	30	5	30	ns
P36-P53	Propagation delay time (t_{PHL}), C_L = 50 pF	X	S	5	21	5	21	5	23	5	23	ns
P54-P88	Propagation delay time (t_{PHL}), C_L = 150 pF	X	Q	5	38	5	38	5	40	5	40	ns
P89-P106	Propagation delay time (t_{PHL}), C_L = 150 pF	X	Q	5	31	5	31	5	33	5	33	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q	Power supply filter capacitance	XI	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
R1-R13	Output crosstalk, high level	XII	Q	0	400	0	400	0	400	0	400	mV
S1-S13	Output crosstalk, low level	XII	Q	0	400	0	400	0	400	0	400	mV
T	Power supply transient amplitude	XIII	Q	0	400	0	400	0	400	0	400	mV
U1-U23	Input capacitance	XIV	Q	2	15	2	15	2	15	2	15	pF
V1-V18	Functional test	XV	X	Input vectors shall produce output patterns as specified in table XV.								

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

Source: https://assist.dla.mil -- Downloaded: 20261[illegible]-20T[illegible]5:38Z
Check the source to verify that this is the current version before use.

* 3.10.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level - Input voltage shall be 0.7 V ±1 percent.
High level - Input voltage shall be 2.0 V ±1 percent.

3.10.3 Output requirements. The output requirements for test patterns shall be as follows:

Low level - Output voltage shall be 0 V minimum, 0.4 V maximum.
High level - Output voltage shall be 2.4 V minimum, V_{CC} maximum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, C, or J are part of a sequence of tests, the order of testing shall be as follows: A, B, C, J where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T and V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Q, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T and V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:38Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/39-1 supersedes part number M28787/39. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation		644 mW
Supply voltage		5.0 V dc ±10 percent
Input voltage		
Low level		0 V minimum, 0.7 V maximum
High level		2.0 V minimum, V_{CC} maximum
Input current (A and B inputs)		
Low level	(V_{IL} = 0.0 V)	-0.88 mA maximum
	(V_{IL} = 0.4 V)	-0.8 mA maximum
High level	(V_{IH} = 2.4 V)	40 µA maximum
	(V_{IH} = 5.5 V)	200 µA maximum
Input current (C1 inputs)		
Low level	(V_{IL} = 0.0 V)	-0.44 mA maximum
	(V_{IL} = 0.4 V)	-0.4 mA maximum
High level	(V_{IH} = 2.4 V)	20 µA maximum
	(V_{IH} = 5.5 V)	100 µA maximum
Output voltage		
Low level		0 V minimum, 0.4 V maximum
High level		2.4 V minimum, V_{CC} maximum
Output drive current maximum capability		
Low level	(V_{OL} = 0.4 V)	4.0 mA
High level	(V_{OH} = 2.4 V)	-400 µA
Propagation delay time (t_{PLH} and t_{PHL})		
A, B, C1 to S	(C_L = 50 pF)	5 ns minimum, 30 ns maximum
	(C_L = 150 pF)	5 ns minimum, 40 ns maximum
A, B, C1 to C2	(C_L = 50 pF)	5 ns minimum, 30 ns maximum
	(C_L = 150 pF)	5 ns minimum, 40 ns maximum
A, B, C1 to C4	(C_L = 50 pF)	5 ns minimum, 23 ns maximum
	(C_L = 150 pF)	5 ns minimum, 33 ns maximum

* 6.5 Module function. The adder module contains one independent 2-bit low-power Schottky adder and two independent 4-bit low-power Schottky adders. The 2-bit adder performs the addition of two 2-bit binary numbers and the sum output for each bit and a carry output. The 4-bit adders perform the addition of two 4-bit binary numbers and provide the sum outputs for each bit and a carry output.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:38Z
Check the source to verify that this is the current version before use.

6.5.1 Input to output relationship. Each adder has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N410)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE II. Input to output relationship. 1/ - 3/

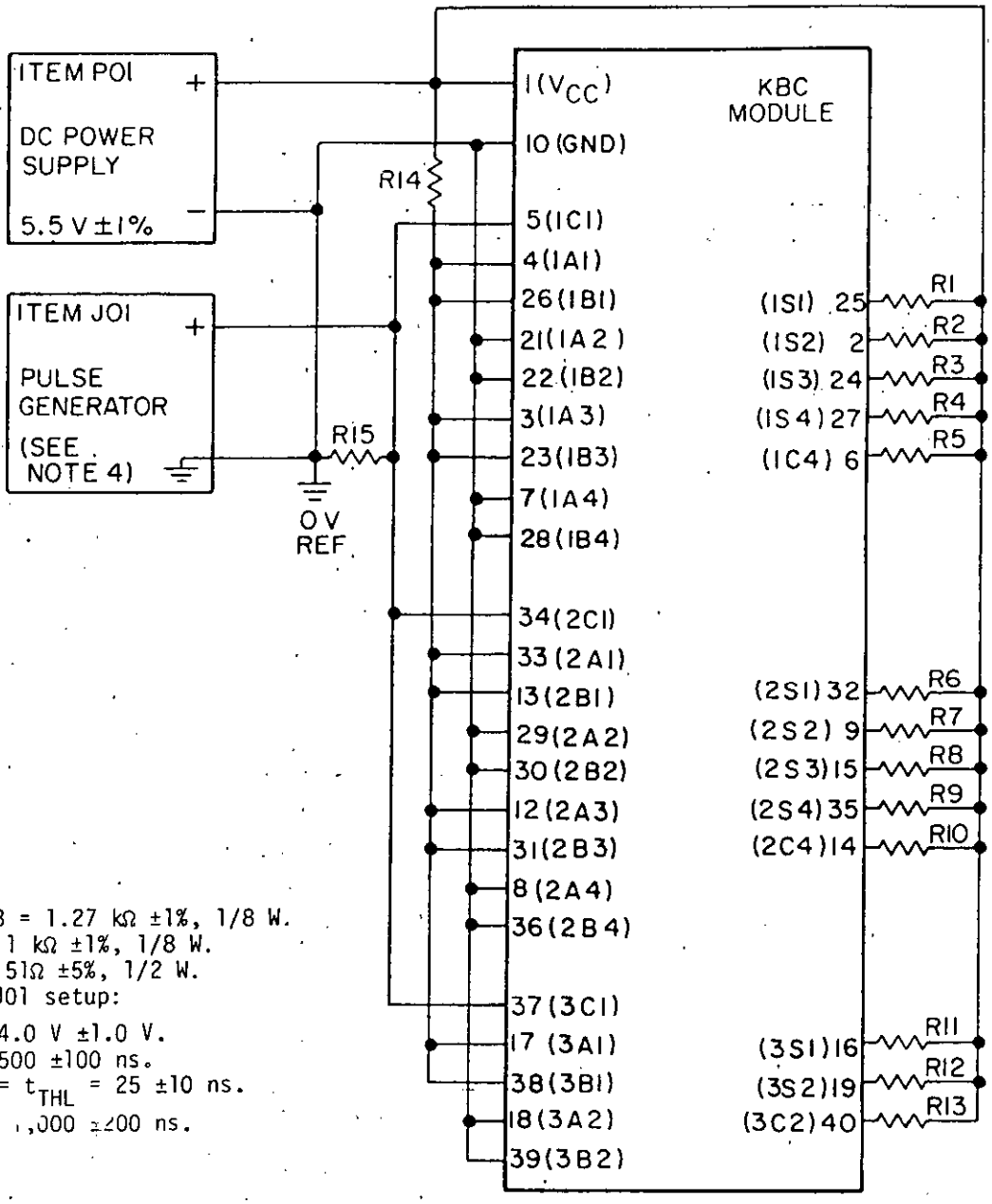
4-bit adder											2-bit adder									
Inputs				Outputs							Inputs				Outputs					
				C1 = L, C2 = L				C1 = H, C2 = H							C1 = L			C1 = H		
A1 / A3	B1 / B3	A2 / A4	B2 / B4	S1 / S3	S2 / S4	C2 / C4	S1 / S3	S2 / S4	C2 / C4		A1	B1	A2	B2	S1	S2	C2	S1	S2	C2
L	L	L	L	L	L	L	H	L	L		L	L	L	L	L	L	L	H	L	L
H	L	L	L	H	L	L	L	H	L		H	L	L	L	H	L	L	L	H	L
L	H	L	L	H	L	L	L	H	L		L	H	L	L	H	L	L	L	H	L
H	H	L	L	L	H	L	H	H	L		H	H	L	L	L	H	L	H	H	L
L	L	H	L	L	H	L	H	H	L		L	L	H	L	L	H	L	H	L	H
H	L	H	L	H	H	L	L	L	H		H	L	H	L	H	H	L	L	L	H
L	H	H	L	H	H	L	L	L	H		L	H	H	L	H	H	L	L	L	H
H	H	H	L	L	L	H	H	L	H		H	H	H	L	L	L	H	H	L	H
L	L	L	H	L	L	H	H	L	H		L	L	L	H	L	L	H	H	L	H
H	L	L	H	H	L	H	L	L	H		H	L	L	H	H	L	H	L	L	H
L	H	L	H	H	L	H	L	L	H		L	H	L	H	H	L	H	L	L	H
H	H	L	H	L	H	H	H	L	H		H	H	L	H	L	H	H	H	L	H
L	L	H	H	L	H	H	H	L	H		L	L	H	H	L	H	H	H	L	H
H	L	H	H	H	H	H	L	H	H		H	L	H	H	H	H	H	L	H	H
L	H	H	H	H	H	H	L	H	H		L	H	H	H	H	H	H	L	H	H
H	H	H	H	L	L	L	H	H	H		H	H	H	H	L	L	L	H	H	H

1/ Input conditions at A1, A2, B1, B2, and C1 are used to determine outputs S1 and S2 and the value of the internal carry C2. The values at C2, A3, B3, A4, and B4 are then used to determine outputs S3, S4, and C4.

2/ L = Low-level.

3/ H = High-level.

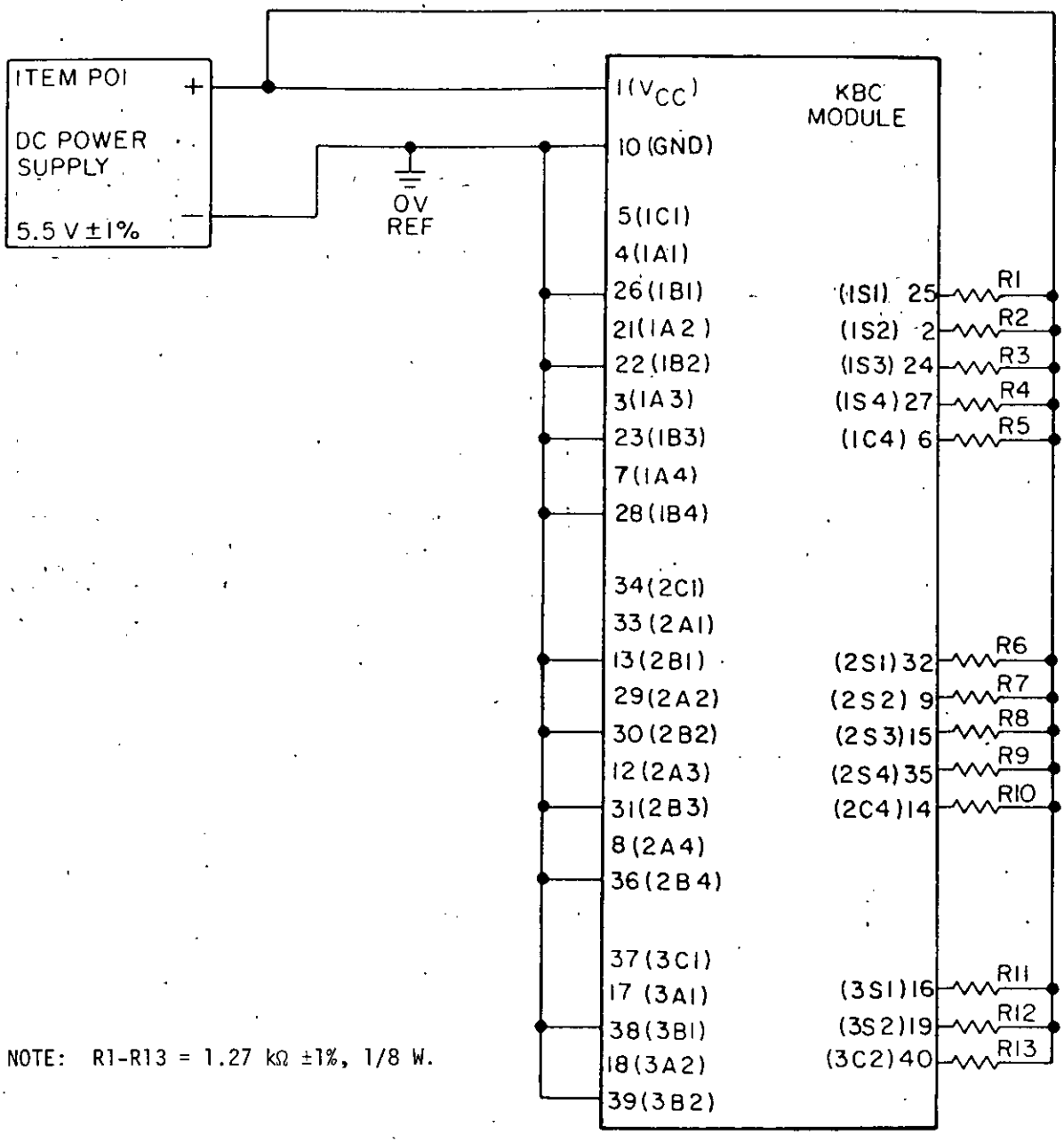
Source: https://assist.dla.mil -- Downloaded: 2026-0-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R13 = 1.27 kΩ ±1%, 1/8 W.
2. R14 = 1 kΩ ±1%, 1/8 W.
3. R15 = 51Ω ±5%, 1/2 W.
4. Item J01 setup:
   PA = 4.0 V ±1.0 V.
   PW = 500 ±100 ns.
   $t_{} = t_{THL}$ = 25 ±10 ns.
   PRF = 1,000 ±200 ns.

* FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: R1-R13 = 1.27 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

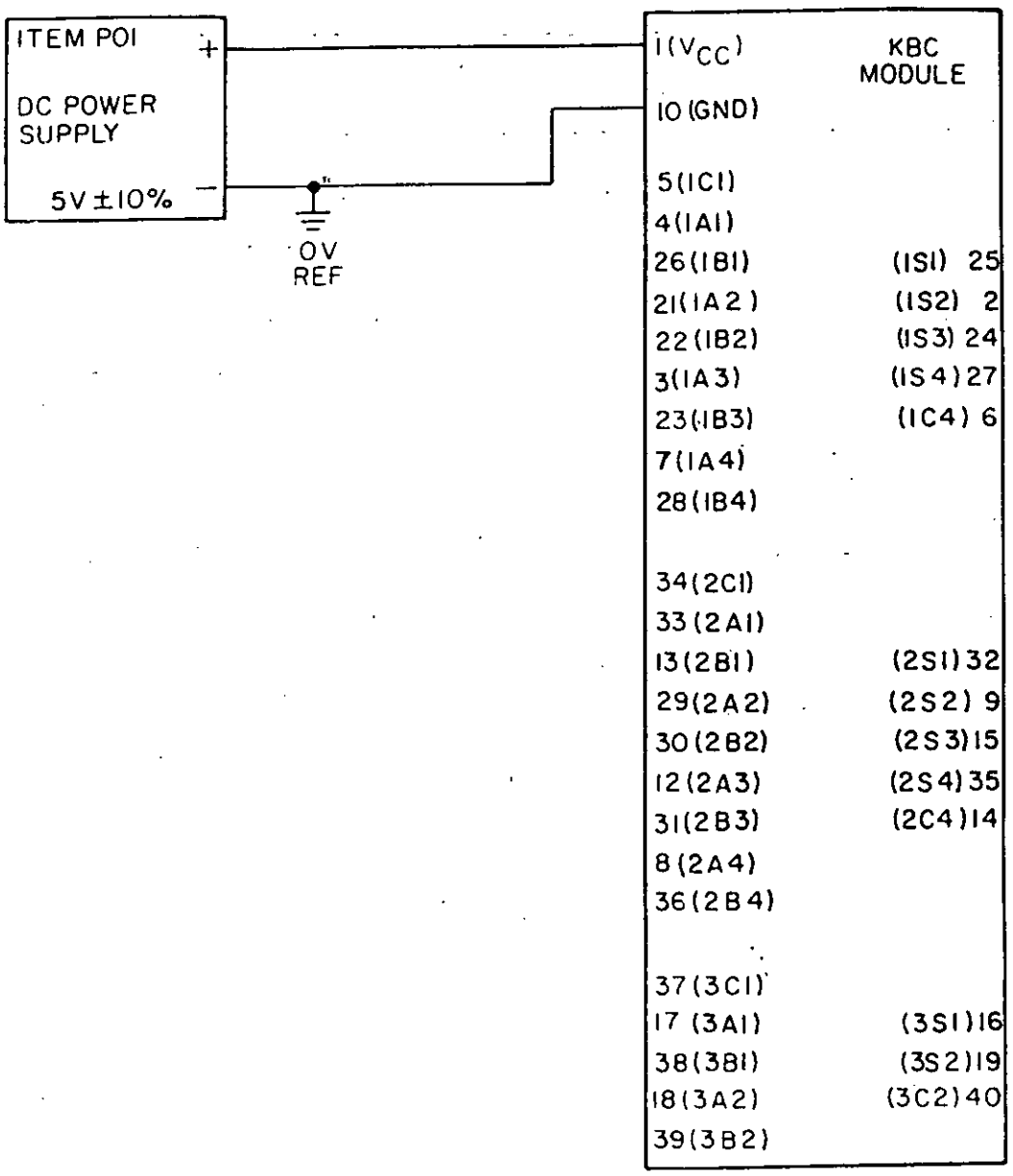


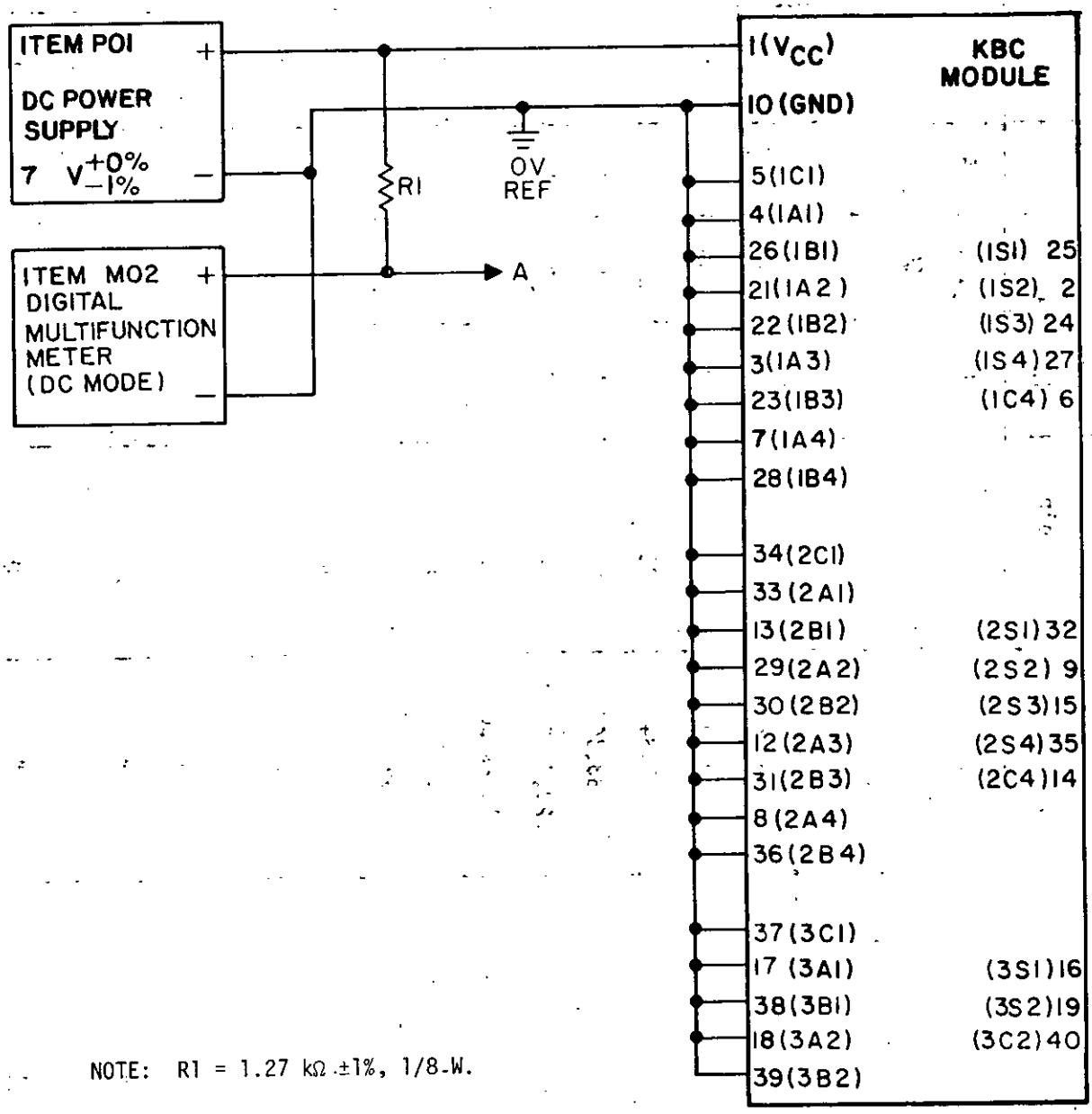
FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

* TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
A1	1S1	V_{OUT} (low loaded)	25	10	800	mV	5
A2	1S2	"	2	"	"	"	"
A3	1S3	"	24	"	"	"	"
A4	1S4	"	27	"	"	"	"
A5	2S1	"	32	"	"	"	"
A6	2S2	"	9	"	"	"	"
A7	2S3	"	15	"	"	"	"
A8	2S4	"	35	"	"	"	"
A9	3S1	"	16	"	"	"	"
A10	3S2	"	19	"	"	"	"
A11	1C4	"	6	"	"	"	"
A12	2C4	"	14	"	"	"	"
A13	3C2	"	40	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026[illegible]-20T[illegible]:38Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 1.27 kΩ ±1%, 1/8-W.

* FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE IV. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	1C1	Input breakdown current	4,26	5	0	100	µA	6
B2	1A1	"	5,26	4	"	200	"	"
B3	1B1	"	4,5	26	"	"	"	"
B4	1A3	"	21,22,23	3	"	"	"	"
B5	1B3	"	3,21,22	23	"	"	"	"
B6	2C1	"	13,33	34	"	100	"	"
B7	2A1	"	13,34	33	"	200	"	"
B8	2B1	"	33,34	13	"	"	"	"
B9	2A3	"	29,30,31	12	"	"	"	"
B10	2B3	"	12,29,30	31	"	"	"	"
B11	3C1	"	17,38	37	"	100	"	"
B12	3A1	"	37,38	17	"	200	"	"
B13	3B1	"	17,37	38	"	"	"	"
B14	1A2	"	Open	21	"	"	"	"
B15	1B2	"	"	22	"	"	"	"
B16	1A4	"	"	7	"	"	"	"
B17	1B4	"	"	28	"	"	"	"
B18	2A2	"	"	29	"	"	"	"
B19	2B2	"	"	30	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-0-20T15:38Z
Check the source to verify that this is the current version before use.

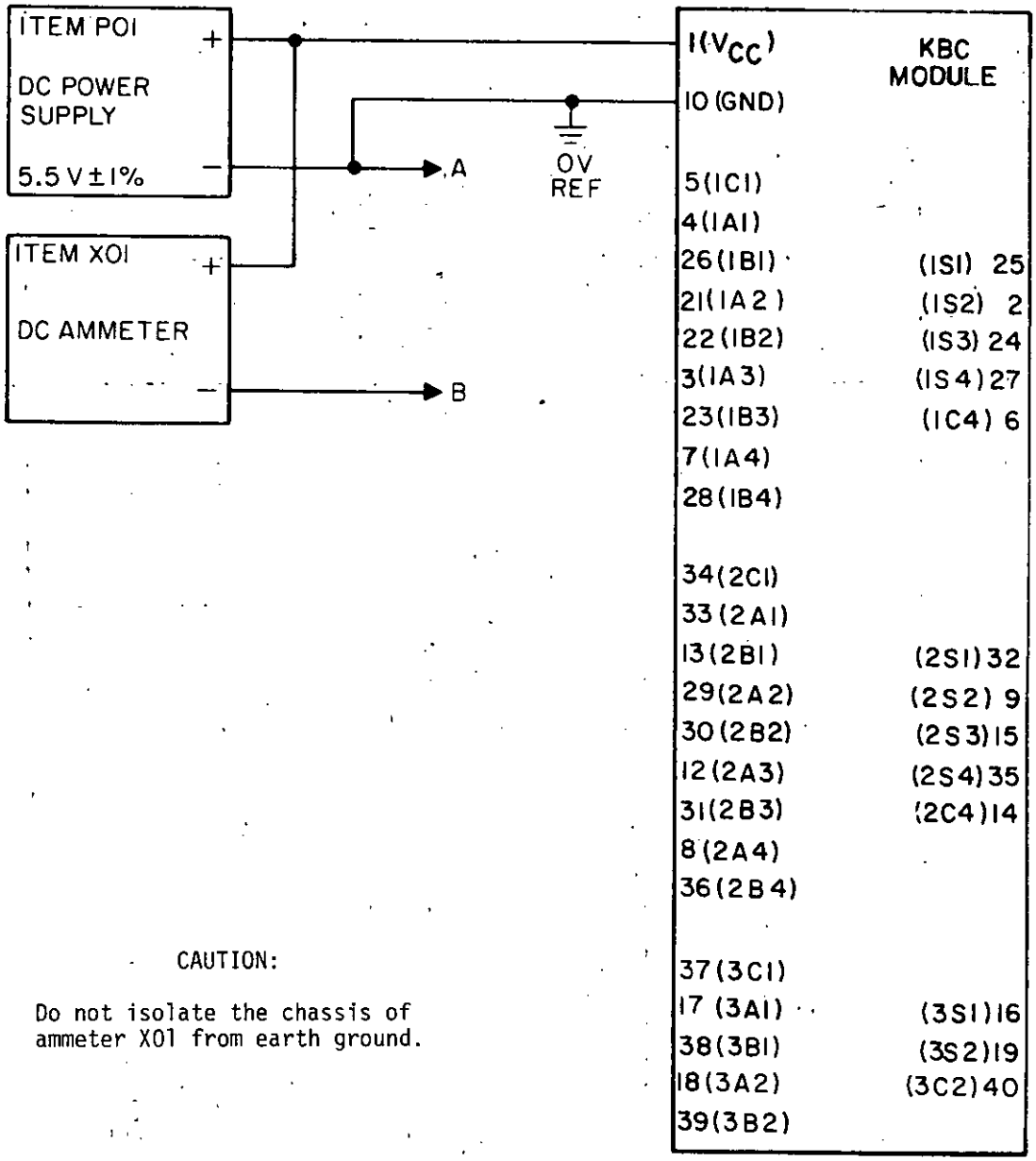
TABLE IV. Input breakdown current test - Continued.

Test condition		Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B20	2A4	Input breakdown current	Open	8	0	200	µA	6
B21	2B4	"	"	36	"	"	"	"
B22	3A2	"	"	18	"	"	"	"
B23	3B2	"	"	39	"	"	"	"

TABLE V. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation, frame	C,11,20	Frame	10	∞	MΩ	7
C2	Isolation, used pins to unused pins	C	11,20	"	"	"	"
C3	Isolation, unused pin to unused pin	11	20	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

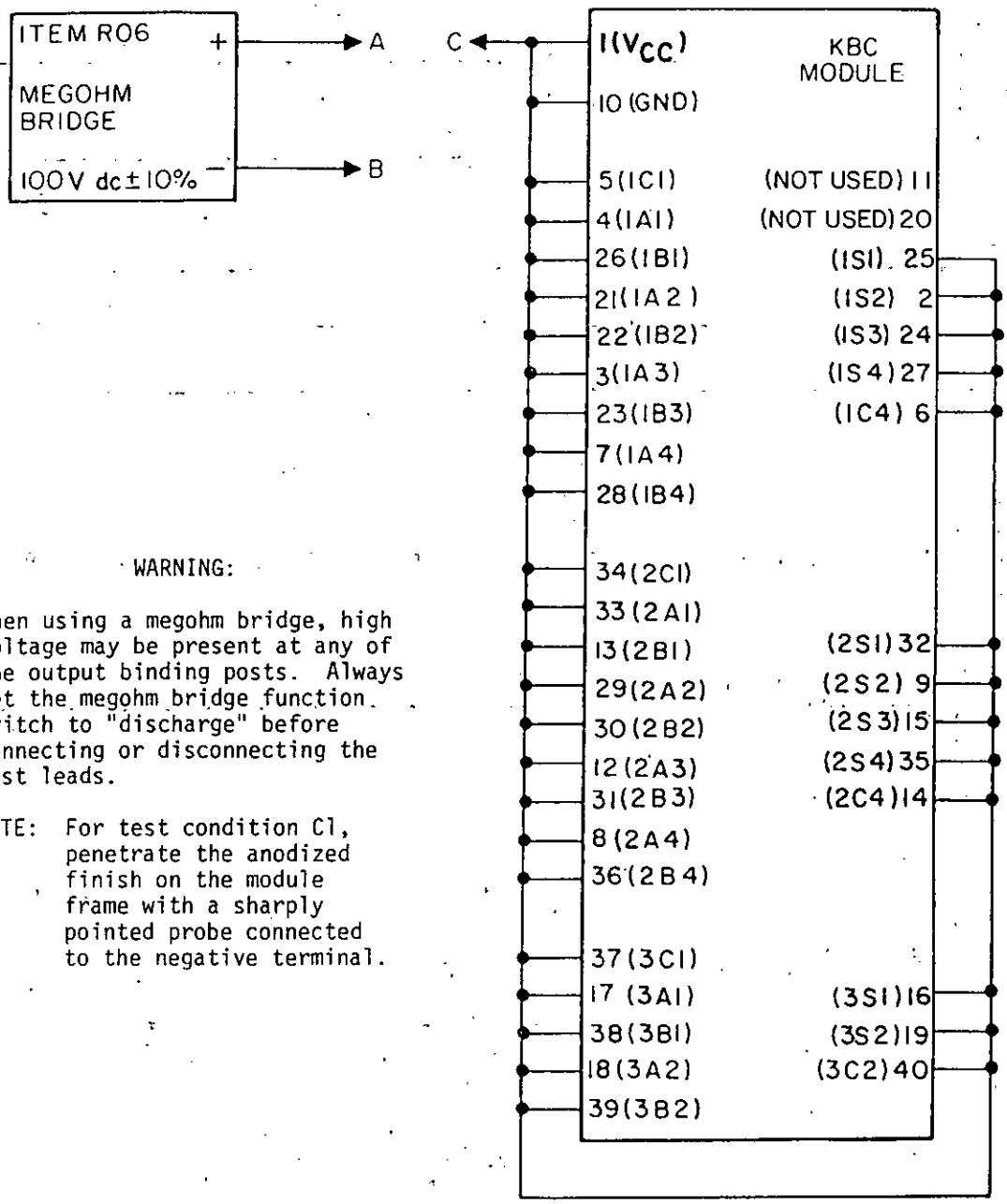


CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 6. Input breakdown current test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition C1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Input clamp diode voltage test.

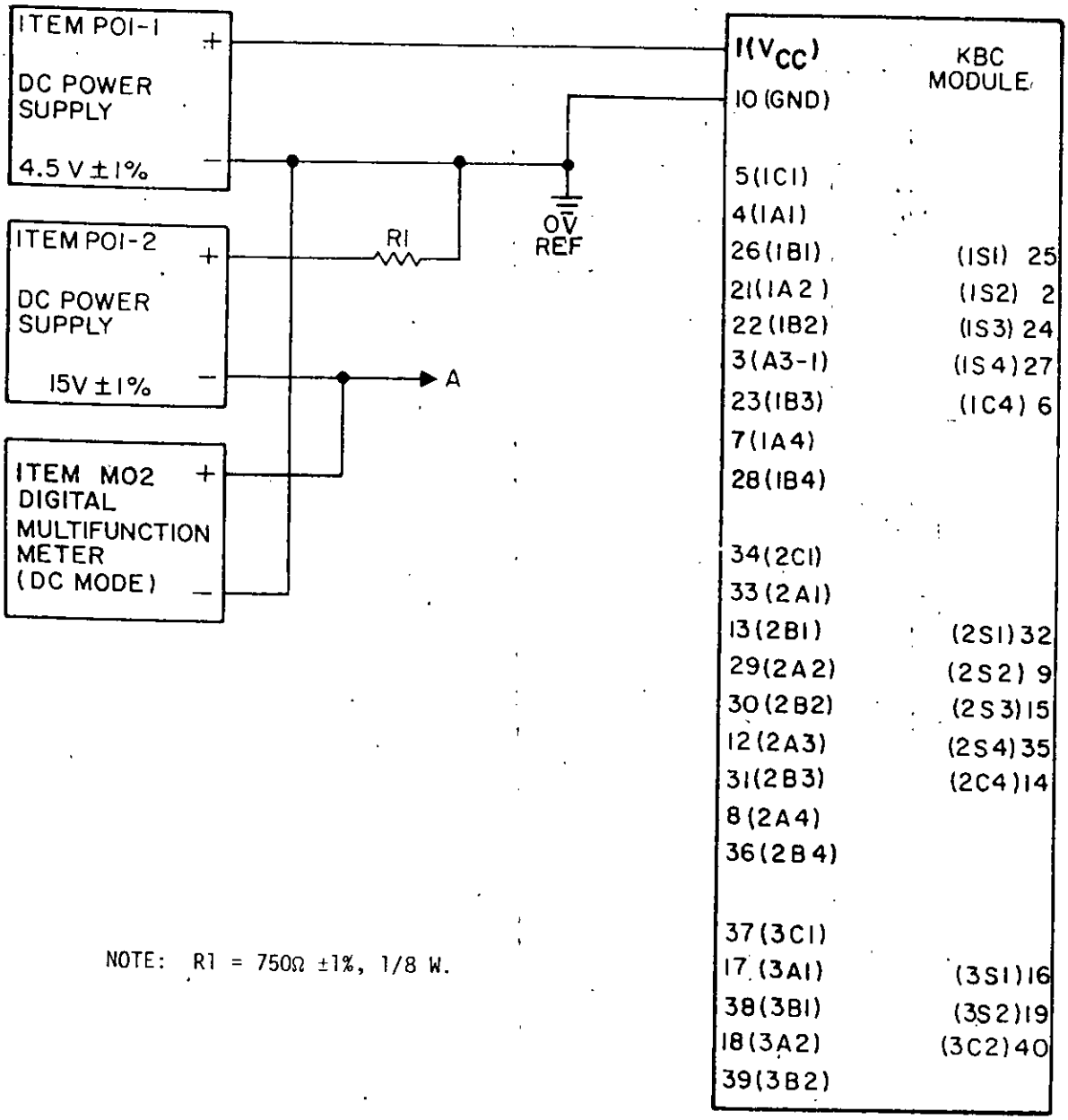
Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
D1	1C1	Input clamp diode voltage	5	-0.4	-1.5	V	8
D2	1A1	"	4	"	"	"	"
D3	1B1	"	26	"	"	"	"
D4	1A2	"	21	"	"	"	"
D5	1B2	"	22	"	"	"	"
D6	1A3	"	3	"	"	"	"
D7	1B3	"	23	"	"	"	"
D8	1A4	"	7	"	"	"	"
D9	1B4	"	28	"	"	"	"
D10	2C1	"	34	"	"	"	"
D11	2A1	"	33	"	"	"	"
D12	2B1	"	13	"	"	"	"
D13	2A2	"	29	"	"	"	"
D14	2B2	"	30	"	"	"	"
D15	2A3	"	12	"	"	"	"
D16	2B3	"	31	"	"	"	"
D17	2A4	"	8	"	"	"	"
D18	2B4	"	36	"	"	"	"
D19	3C1	"	37	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20261 0-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VI. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
D20	3A1	Input clamp diode voltage	17	-0.4	-1.5	V	8
D21	3B1	"	38	"	"	"	"
D22	3A2	"	18	"	"	"	"
D23	3B2	"	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 750Ω ±1%, 1/8 W.

* FIGURE 8. Input clamp diode voltage test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

* TABLE VII. Output voltage (low-level and high-level) test.

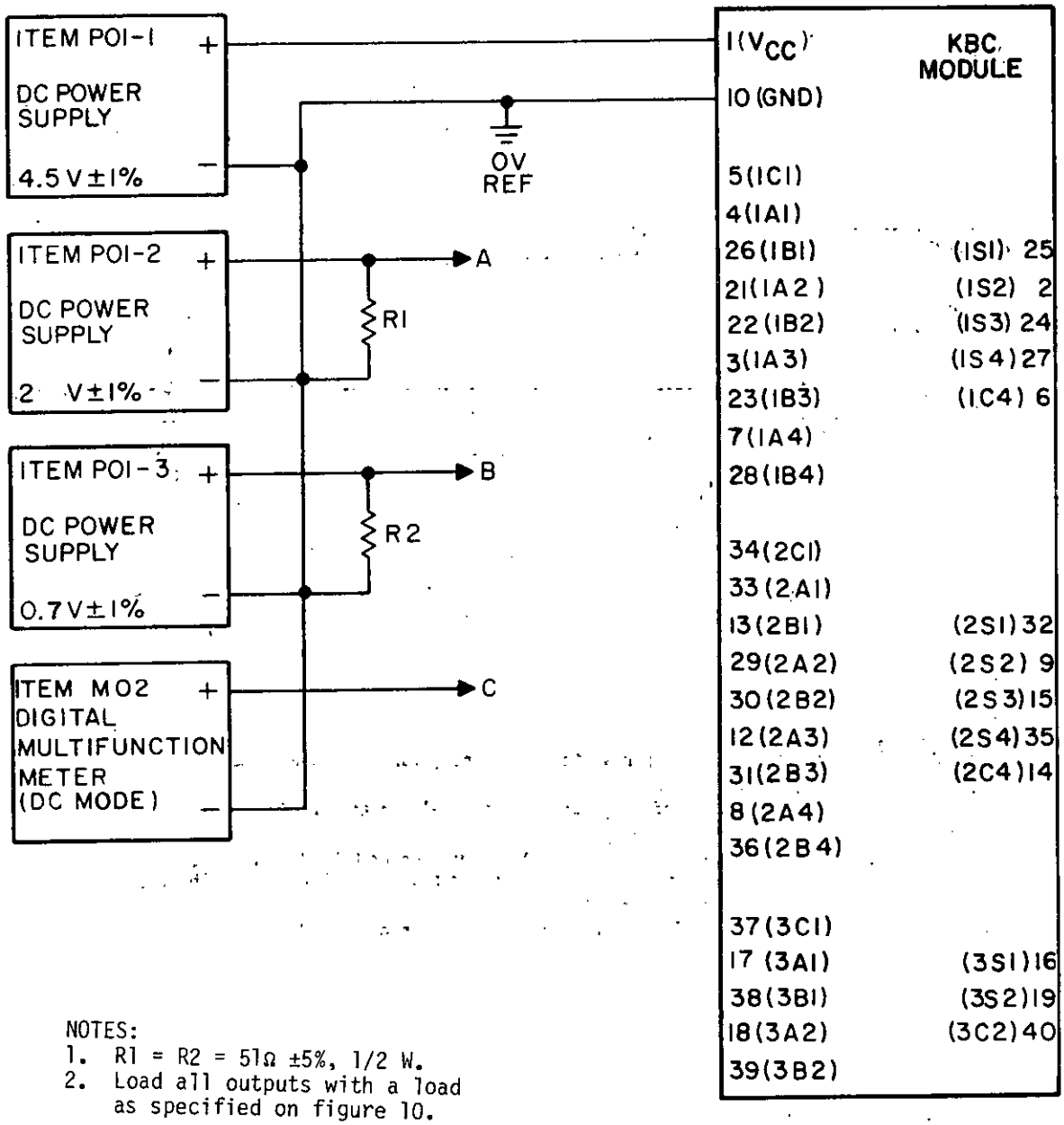
Test condition	Parameter		Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
E1	1S1	Output voltage, low level	Open	3,4,5,7,21, 22,23,26,28	25	0	400	mV	9
E2	1S2	"	"	"	2	"	"	"	"
E3	1S3	"	"	"	24	"	"	"	"
E4	1S4	"	"	"	27	"	"	"	"
E5	1C4	"	"	"	6	"	"	"	"
E6	2S1	"	"	8,12,13,29, 30,31,33, 34,36	32	"	"	"	"
E7	2S2	"	"	"	9	"	"	"	"
E8	2S3	"	"	"	15	"	"	"	"
E9	2S4	"	"	"	35	"	"	"	"
E10	2C4	"	"	"	14	"	"	"	"
E11	3S1	"	"	17,18,37, 38,39	16	"	"	"	"
E12	3S2	"	"	"	19	"	"	"	"
E13	3C2	"	"	"	40	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

* TABLE VII. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	1S1	Output voltage, high level	3,4,5,7,21, 22,23,26,28	Open	25	2.4	4.5	V	9
F2	1S2	"	"	"	2	"	"	"	"
F3	1S3	"	"	"	24	"	"	"	"
F4	1S4	"	"	"	27	"	"	"	"
F5	1C4	"	"	"	6	"	"	"	"
F6	2S1	"	8,12,13,29, 30,31,33, 34,36	"	32	"	"	"	"
F7	2S2	"	"	"	9	"	"	"	"
F8	2S3	"	"	"	15	"	"	"	"
F9	2S4	"	"	"	35	"	"	"	"
F10	2C4	"	"	"	14	"	"	"	"
F11	3S1	"	17,18,37, 38,39	"	16	"	"	"	"
F12	3S2	"	"	"	19	"	"	"	"
F13	3C2	"	"	"	40	"	"	"	"

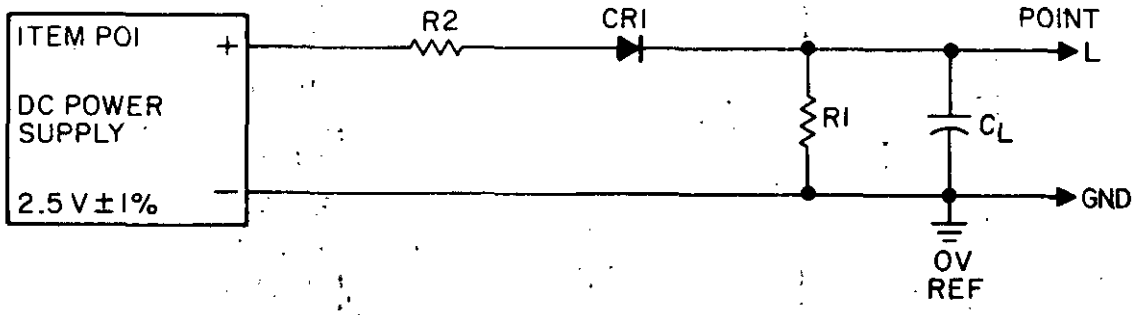
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a load as specified on figure 10.

* FIGURE 9. Output voltage (low-level and high-level) test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:

1. CR1 = 1N4150, or equivalent.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions L1-L13, M1-M13, N1-N53, P1-P53, R and S.
3. C_L = 150 ±5 pF (including probe and parasitic capactiance) for test conditions L14-L26, M14-M26, N54-N106, and P54-P106.
4. R1 = 6.04 kΩ ±1%, 1/8 W (current flow of -400 µA ±5% with point L held at 2.4 V).
5. R2 = Select value for current flow of 4.0 mA ±5% out of point L with point L held at 0.4 V.
6. C_L is not required for dc electrical tests E and F.
7. During output switching, item POI shall have a maximum overshoot of 500 mV.

FIGURE 10. KBC logic load circuit.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current (low-level and high-level) and output short circuit current test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	1C1	Input current, low level	Open	5	Open	-0.01	-0.44	mA	11
G2	1A1	"	"	4	"	"	-0.88	"	"
G3	1B1	"	"	26	"	"	"	"	"
G4	1A3	"	"	3	"	"	"	"	"
G5	1B3	"	"	23	"	"	"	"	"
G6	2C1	"	"	34	"	"	-0.44	"	"
G7	2A1	"	"	33	"	"	-0.88	"	"
G8	2B1	"	"	13	"	"	"	"	"
G9	2A3	"	"	12	"	"	"	"	"
G10	2B3	"	"	31	"	"	"	"	"
G11	3C1	"	"	37	"	"	-0.44	"	"
G12	3A1	"	"	17	"	"	-0.88	"	"
G13	3B1	"	"	38	"	"	"	"	"
G14	1A2	"	"	21	"	"	"	"	"
G15	1B2	"	"	22	"	"	"	"	"
G16	1A4	"	"	7	"	"	"	"	"
G17	1B4	"	"	28	"	"	"	"	"
G18	2A2	"	"	29	"	"	"	"	"
G19	2B2	"	"	30	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G20	2A4	Input current, low level	Open	8	Open	-0.01	-0.88	mA	11
G21	2B4	"	"	36	"	"	"	"	"
G22	3A2	"	"	18	"	"	"	"	"
G23	3B2	"	"	39	"	"	"	"	"
H1	1C1	Input current, high level	5	Open	4,26	0	20	µA	"
H2	1A1	"	4	"	5,26	"	40	"	"
H3	1B1	"	26	"	4,5	"	"	"	"
H4	1A3	"	3	"	21,22,23	"	"	"	"
H5	1B3	"	23	"	3,21,22	"	"	"	"
H6	2C1	"	34	"	13,33	"	20	"	"
H7	2A1	"	33	"	13,34	"	40	"	"
H8	2B1	"	13	"	33,34	"	"	"	"
H9	2A3	"	12	"	29,30,31	"	"	"	"
H10	2B3	"	31	"	12,29,30	"	"	"	"
H11	3C1	"	37	"	17,38	"	20	"	"
H12	3A1	"	17	"	37,38	"	40	"	"
H13	3B1	"	38	"	17,37	"	"	"	"
H14	1A2	"	21	"	Open	"	"	"	"
H15	1B2	"	22	"	Open	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition		Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H16	1A4	Input current, high level	7	Open	Open	0	40	µA	11
H17	1B4	"	28	"	"	"	"	"	"
H18	2A2	"	29	"	"	"	"	"	"
H19	2B2	"	30	"	"	"	"	"	"
H20	2A4	"	8	"	"	"	"	"	"
H21	2B4	"	36	"	"	"	"	"	"
H22	3A2	"	18	"	"	"	"	"	"
H23	3B2	"	39	"	"	"	"	"	"
J1	1S1	Output short circuit current	Open	25	"	-6	-120	mA	"
J2	1S2	"	"	2	"	"	"	"	"
J3	1S3	"	"	24	"	"	"	"	"
J4	1S4	"	"	27	"	"	"	"	"
J5	1C4	"	"	6	"	"	"	"	"
J6	2S1	"	"	32	"	"	"	"	"
J7	2S2	"	"	9	"	"	"	"	"
J8	2S3	"	"	15	"	"	"	"	"
J9	2S4	"	"	35	"	"	"	"	"
J10	2C4	"	"	14	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.

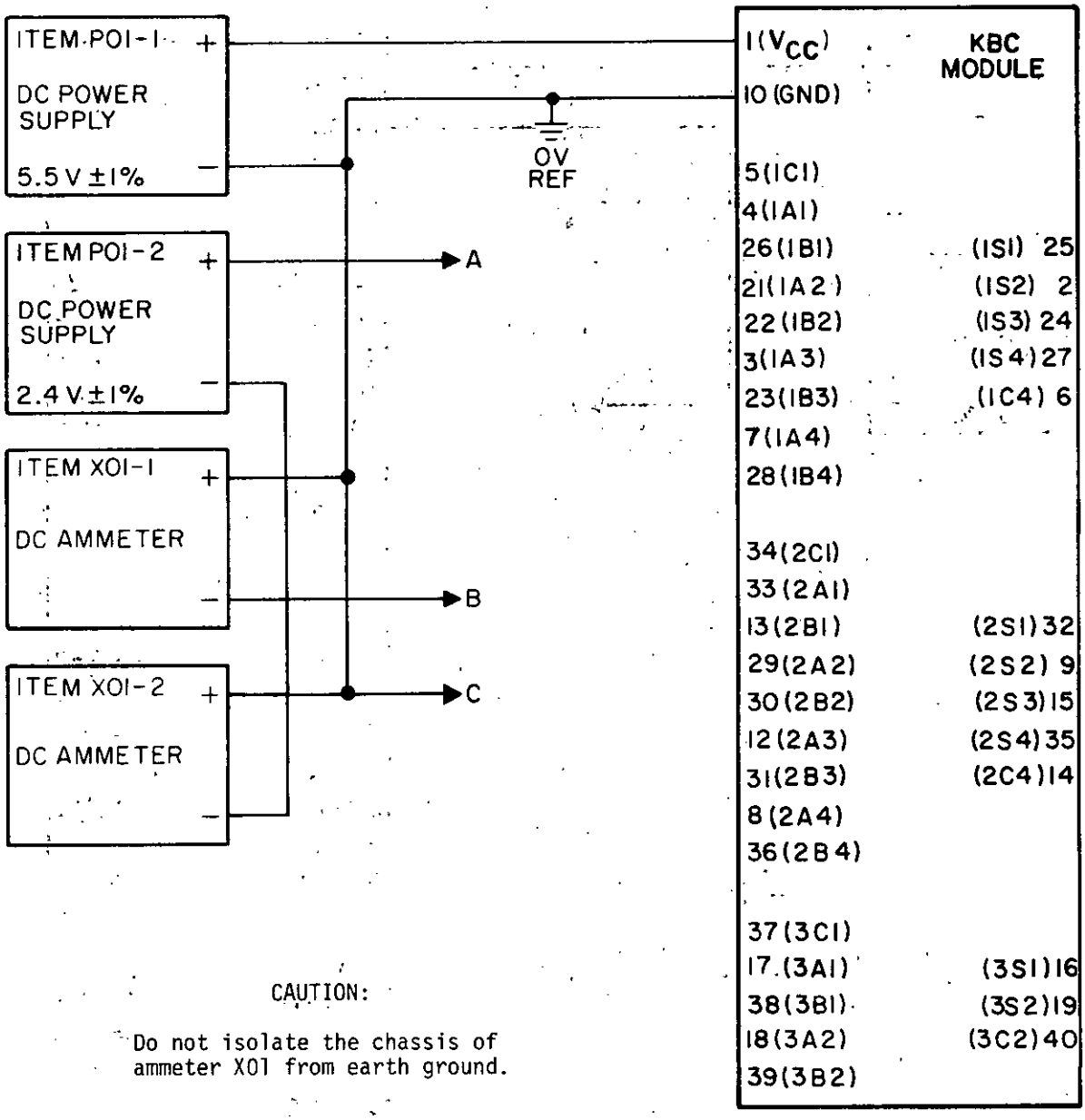
TABLE VIII. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J11	3S1	Output short circuit current	Open	16	Open	-6	-120	mA	11
J12	3S2	"	"	19	"	"	"	"	"
J13	3C2	"	"	40	"	"	"	"	"

TABLE IX. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K1	Power supply current, low level	13,22,23,26,28, 30,31,36,38,39	10	117	mA	12
K2	Power supply current, high level	3,4,5,7,8,12, 17,18,21,29,33, 34,37	10	102	mA	12

Source: https://assist.dla.mil -- Downloaded: 20261[illegible]-20T[illegible]:38Z
Check the source to verify that this is the currentversion before use.



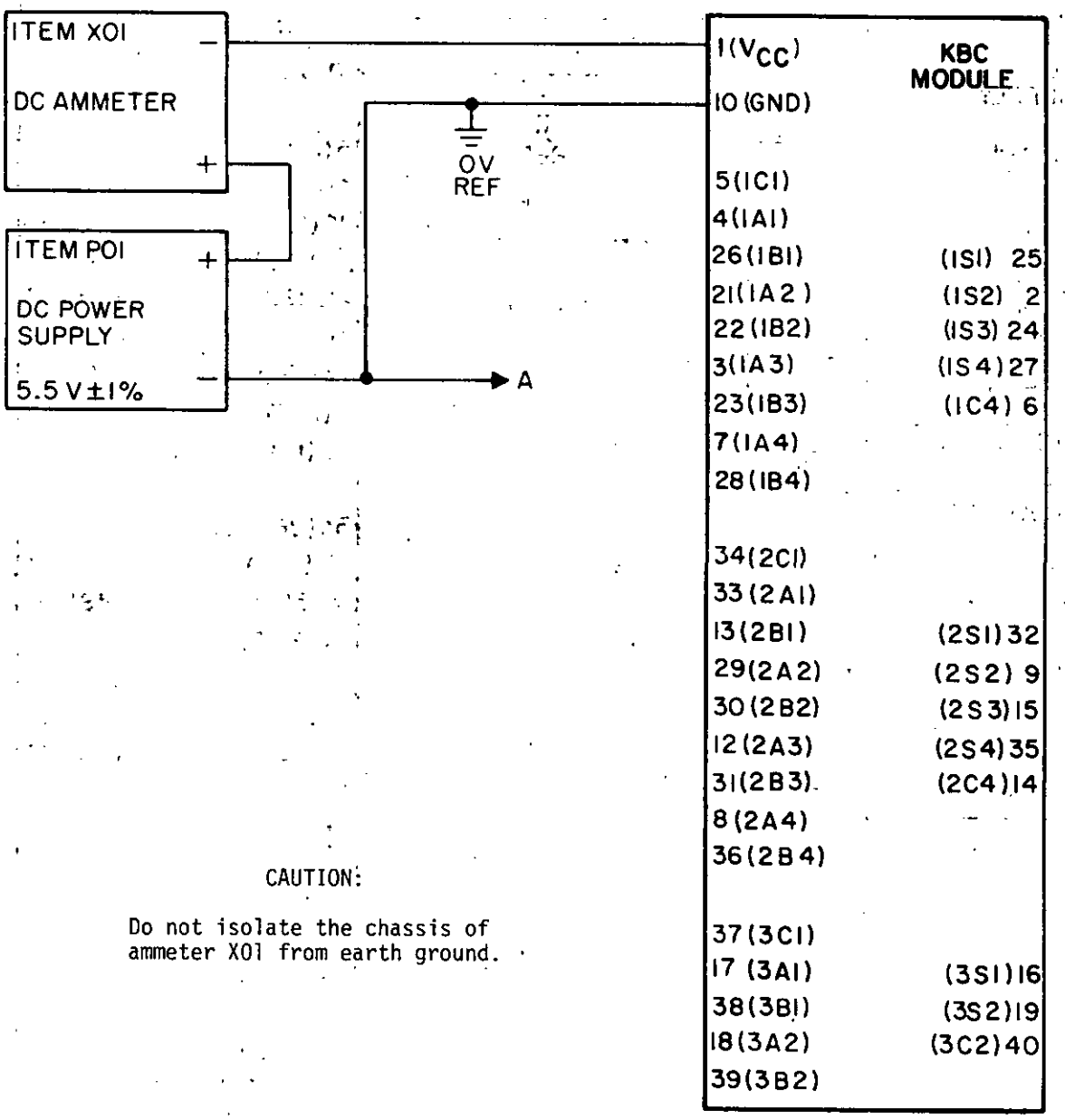
CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:
1. Since items XOI-1 and XOI-2 are not used simultaneously, only one item XOI is required.
2. With point A disconnected, item XOI-1 should indicate zero current on the 1 µA range.
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

*FIGURE 11. Input current (low-level and high-level) and output short circuit current test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

FIGURE 12. Power supply current (low-level and high-level) test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1S1 Transition time (t_{TLH}), C_L = 50 pF	4	5	25	21,22,26	5	50	ns	13,14
L2	1S2 "	4,21	"	2	22,26	"	"	"	"
L3	1S3 "	3,4,21	"	24	22,23,26	"	"	"	"
L4	1S4 "	3,4,7,21	"	27	22,23,26, 28	"	"	"	"
L5	1C4 "	3,4,7,21	"	6	22,23,26, 28	"	"	"	"
L6	2S1 "	33	34	32	13,29,30	"	"	"	"
L7	2S2 "	29,33	"	9	13,30	"	"	"	"
L8	2S3 "	12,29,33	"	15	13,30,31	"	"	"	"
L9	2S4 "	8,12,29, 33	"	35	13,30,31, 36	"	"	"	"
L10	2C4 "	8,12,29, 33	"	14	13,30,31, 36	"	"	"	"
L11	3S1 "	17	37	16	38	"	"	"	"
L12	3S2 "	17,18	"	19	38,39	"	"	"	"
L13	3C2 "	17,18	"	40	38,39	"	"	"	"
L14	1S1 Transition time (t_{TLH}), C_L = 150 pF	4	5	25	21,22,26	10	110	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L15	1S2 Transition time (t_{TLH}), C_L = 150 pF	4,21	5	2	22,26	10	110	ns	13,14
L16	1S3 "	3,4,21	"	24	22,23,26	"	"	"	"
L17	1S4 "	3,4,7,21	"	27	22,23,26, 28	"	"	"	"
L18	1C4 "	3,4,7,21	"	6	22,23,26, 28	"	"	"	"
L19	2S1 "	33	34	32	13,29,30	"	"	"	"
L20	2S2 "	29,33	"	9	13,30	"	"	"	"
L21	2S3 "	12,29,33	"	15	13,30,31	"	"	"	"
L22	2S4 "	8,12,29, 33	"	35	13,30,31, 36	"	"	"	"
L23	2C4 "	8,12,29, 33	"	14	13,30,31, 36	"	"	"	"
L24	3S1 "	17	37	16	38	"	"	"	"
L25	3S2 "	17,18	"	19	38,39	"	"	"	"
L26	3C2 "	17,18	"	40	38,39	"	"	"	"
M1	1S1 Transition time (t_{THL}), C_L = 50 pF	4	5	25	21,22,26	3	25	"	"
M2	1S2 Transition time (t_{THL}), C_L = 50 pF	4,21	5	2	22,26	3	25	"	"

Source: https://assist.dla.mil -- Downloaded: 20260120T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M3	1S3 Transition time (t_{THL}), C_L = 50 pF	3,4,21	5	24	22,23,26	3	25	ns	13,14
M4	1S4 "	3,4,7,21	"	27	22,23,26, 28	"	"	"	"
M5	1C4 "	3,4,7,21	"	6	22,23,26, 28	"	"	"	"
M6	2S1 "	33	34	32	13,29,30	"	"	"	"
M7	2S2 "	29,33	"	9	13,30	"	"	"	"
M8	2S3 "	12,29,33	"	15	13,30,31	"	"	"	"
M9	2S4 "	8,12,29, 33	"	35	13,30,31, 36	"	"	"	"
M10	2C4 "	8,12,29, 33	"	14	13,30,31, 36	"	"	"	"
M11	3S1 "	17	37	16	38	"	"	"	"
M12	3S2 "	17,18	"	19	38,39	"	"	"	"
M13	3C2 "	17,18	"	40	38,39	"	"	"	"
M14	1S1 Transition time (t_{THL}), C_L = 150 pF	4	5	25	21,22,26	6	35	"	"
M15	1S2 "	4,21	"	2	22,26	"	"	"	"
M16	1S3 "	3,4,21	"	24	22,23,26	"	"	"	"
M17	1S4 "	3,4,7,21	"	27	22,23,26, 28	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M18	1C4 Transition time (t_{THL}), C_L = 150 pF	3,4,7,21	5	6	22,23,26, 28	6	35	ns	13,14
M19	2S1 "	33	34	32	13,29,30	"	"	"	"
M20	2S2 "	29,33	"	9	13,30	"	"	"	"
M21	2S3 "	12,33,39	"	15	13,30,31	"	"	"	"
M22	2S4 "	8,12,29, 33	"	35	13,30,31, 36	"	"	"	"
M23	2C4 "	8,12,29, 33	"	14	13,30,31, 36	"	"	"	"
M24	3S1 "	17	37	16	38	"	"	"	"
M25	3S2 "	17,18	"	19	38,39	"	"	"	"
M26	3C2 "	17,18	"	40	38,39	"	"	"	"
N1	1C1 to 1S1 Propagation delay time (t_{PLH}), C_L = 50 pF	3,4,7,21	5	25	22,23,26, 28	5	28	"	"
N2	1C1 to 1S2 "	"	"	2	"	"	"	"	"
N3	1C1 to 1S3 "	"	"	24	"	"	"	"	"
N4	1C1 to 1S4 "	"	"	27	"	"	"	"	"
N5	2C1 to 2S1 "	8,12,29, 33	34	32	13,30,31, 36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N6	2C1 to 2S2	Propagation delay time (t_{PLH}), C_L = 50 pF	8,12,29, 33	34	9	13,30,31, 36	5	28	ns	13,14
N7	2C1 to 2S3	"	8,12,29, 33	"	15	"	"	"	"	"
N8	2C1 to 2S4	"	8,12,29, 33	"	35	"	"	"	"	"
N9	3C1 to 3S1	"	17,18	37	16	38,39	"	"	"	"
N10	3C1 to 3S2	"	17,18	37	19	38,39	"	"	"	"
N11	1A1 to 1S1	"	Open	4	25	3,5,7,21, 22,23,26, 28	"	"	"	"
N12	1A2 to 1S2	"	"	21	2	3,4,5,7, 22,23,26, 28	"	"	"	"
N13	1A3 to 1S3	"	"	3	24	4,5,7,21, 22,23,26, 28	"	"	"	"
N14	1A4 to 1S4	"	"	7	27	3,4,5,21, 22,23,26, 28	"	"	"	"
N15	1B1 to 1S1	"	"	26	25	3,4,5,7, 21,22,23, 28	"	"	"	"
N16	1B2 to 1S2	"	"	22	2	3,4,5,7, 21,23,26, 28	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N17	1B3 to 1S3	Propagation delay time (t_{PLH}), C_L = 50 pF	Open	23	24	3,4,5,7, 21,22,26, 28	5	28	ns	13,14
N18	1B4 to 1S4	"	"	28	27	3,4,5,7, 21,22,23, 26	"	"	"	"
N19	2A1 to 2S1	"	"	33	32	8,12,13, 29,30,31, 34,36	"	"	"	"
N20	2A2 to 2S2	"	"	29	9	8,12,13, 30,31,33, 34,36	"	"	"	"
N21	2A3 to 2S3	"	"	12	15	8,13,29, 30,31,33, 34,36	"	"	"	"
N22	2A4 to 2S4	"	"	8	35	12,13,29, 30,31,33, 34,36	"	"	"	"
N23	2B1 to 2S1	"	"	13	32	8,12,29, 30,31,33, 34,36	"	"	"	"
N24	2B2 to 2S2	"	"	30	9	8,12,13, 29,31,33, 34,36	"	"	"	"
N25	2B3 to 2S3	"	"	31	15	8,12,13, 29,30,33, 34,36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20260420T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N26	2B4 to 2S4	Propagation delay time (t_{PLH}), C_L = 50 pF	Open	36	35	8,12,13, 29,30,31, 33,34	5	28	ns	13,14
N27	3A1 to 3S1	"	"	17	16	18,37,38, 39	"	"	"	"
N28	3A2 to 3S2	"	"	18	19	17,37,38, 39	"	"	"	"
N29	3B1 to 3S1	"	"	38	16	17,18,37, 39	"	"	"	"
N30	3B2 to 3S2	"	"	39	19	17,18,37, 38	"	"	"	"
N31	3A1 to 3C2	"	18,38	17	40	37,39	"	"	"	"
N32	3A2 to 3C2	"	39	18	"	17,37,38	"	"	"	"
N33	3B1 to 3C2	"	17,39	38	"	18,37	"	"	"	"
N34	3B2 to 3C2	"	18	39	"	17,37,38	"	"	"	"
N35	3C1 to 3C2	"	17,18	37	"	38,39	"	"	"	"
N36	1A1 to 1C4	"	7,22,23, 26	4	6	3,5,21,28	"	21	"	"
N37	1A2 to 1C4	"	7,22,23	21	"	3,4,5,26, 28	"	"	"	"
N38	1A3 to 1C4	"	7,23	3	"	4,5,21,22, 26,28	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N39	1A4 to 1C4	Propagation delay time (t_{PLH}), C_L = 50 pF	28	7	6	3,4,5,21, 22,23,26	5	21	ns	13,14
N40	1B1 to 1C4	"	3,4,21, 28	26	"	5,7,22,23	"	"	"	"
N41	1B2 to 1C4	"	3,21,28	22	"	4,5,7,23, 26	"	"	"	"
N42	1B3 to 1C4	"	3,28	23	"	4,5,7,21, 22,26	"	"	"	"
N43	1B4 to 1C4	"	7	28	"	3,4,5,21, 22,23,26	"	"	"	"
N44	2A1 to 2C4	"	8,13,30, 31	33	14	12,29,34, 36	"	"	"	"
N45	2A2 to 2C4	"	8,30,31	29	"	12,13,33, 34,36	"	"	"	"
N46	2A3 to 2C4	"	8,31	12	"	13,29,30, 33,34,36	"	"	"	"
N47	2A4 to 2C4	"	36	8	"	12,13,29, 30,31,33, 34	"	"	"	"
N48	2B1 to 2C4	"	12,29,33, 36	13	"	8,30,31,34	"	"	"	"
N49	2B2 to 2C4	"	12,29,36	30	"	8,13,31, 33,34	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026/10/20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N50	2B3 to 2C4	Propagation delay time (t_{PLH}), C_L = 50 pF	12,36	31	14	8,13,29,30, 33,34	5	21	ns	13,14
N51	2B4 to 2C4	"	8	36	14	12,13,29, 30,31,33, 34	"	"	"	"
N52	1C1 to 1C4	"	3,4,7,21	5	6	22,23,26, 28	"	"	"	"
N53	2C1 to 2C4	"	8,12,29, 33	34	14	13,30,31, 36	"	"	"	"
N54	1C1 to 1S1	Propagation delay time (t_{PLH}), C_L = 150 pF	3,4,7,21	5	25	22,23,26, 28	"	38	"	"
N55	1C1 to 1S2	"	"	"	2	"	"	"	"	"
N56	1C1 to 1S3	"	"	"	24	"	"	"	"	"
N57	1C1 to 1S4	"	"	"	27	"	"	"	"	"
N58	2C1 to 2S1	"	8,12,29, 33	34	32	13,30,31, 36	"	"	"	"
N59	2C1 to 2S2	"	"	"	9	"	"	"	"	"
N60	2C1 to 2S3	"	"	"	15	"	"	"	"	"
N61	2C1 to 2S4	"	"	"	35	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N62	3C1 to 3S1	Propagation delay time (t_{PLH}), C_L = 150 pF	17,18	37	16	38,39	5	38	ns	13,14
N63	3C1 to 3S2	"	17,18	37	19	38,39	"	"	"	"
N64	1A1 to 1S1	"	Open	4	25	3,5,7,21, 22,23,26, 28	"	"	"	"
N65	1A2 to 1S2	"	"	21	2	3,4,5,7, 22,23,26, 28	"	"	"	"
N66	1A3 to 1S3	"	"	3	24	4,5,7,21, 22,23,26, 28	"	"	"	"
N67	1A4 to 1S4	"	"	7	27	3,4,5,21, 22,23,26, 28	"	"	"	"
N68	1B1 to 1S1	"	"	26	25	3,4,5,7, 21,22,23, 28	"	"	"	"
N69	1B2 to 1S2	"	"	22	2	3,4,5,7, 21,23,26, 28	"	"	"	"
N70	1B3 to 1S3	"	"	23	24	3,4,5,7, 21,22,26, 28	"	"	"	"
N71	1B4 to 1S4	"	"	28	27	3,4,5,7, 21,22,23, 26	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N72	2A1 to 2S1 Propagation delay time (t_{PLH}), C_L = 150 pF	Open	33	32	8,12,13, 29,30,31, 34,36	5	38	ns	13,14
N73	2A2 to 2S2 "	"	29	9	8,12,13, 30,31,33, 34,36	"	"	"	"
N74	2A3 to 2S3 "	"	12	15	8,13,29, 30,31,33, 34,36	"	"	"	"
N75	2A4 to 2S4 "	"	8	35	12,13,29, 30,31,33, 34,36	"	"	"	"
N76	2B1 to 2S1 "	"	13	32	8,12,29, 30,31,33, 34,36	"	"	"	"
N77	2B2 to 2S2 "	"	30	9	8,12,13, 29,31,33, 34,36	"	"	"	"
N78	2B3 to 2S3 "	"	31	15	8,12,13, 29,30,33, 34,36	"	"	"	"
N79	2B4 to 2S4 "	"	36	35	8,12,13, 29,30,31, 33,34	"	"	"	"
N80	3A1 to 3S1 "	"	17	16	18,37,38, 39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N81	3A2 to 3S2 Propagation delay time (t_{PLH}), C_L = 150 pF	Open	18	19	17,37,38, 39	5	38	ns	13,14
N82	3B1 to 3S1 "	"	38	16	17,18,37, 39	"	"	"	"
N83	3B2 to 3S2 "	"	39	19	17,18,37, 38	"	"	"	"
N84	3A1 to 3C2 "	18,38	17	40	37,39	"	"	"	"
N85	3A2 to 3C2 "	39	18	"	17,37,38	"	"	"	"
N86	3B1 to 3C2 "	17,39	38	"	18,37	"	"	"	"
N87	3B2 to 3C2 "	18	39	"	17,37,38	"	"	"	"
N88	3C1 to 3C2 "	17,18	37	"	38,39	"	"	"	"
N89	1A1 to 1C4 "	7,22,23, 26	4	6	3,5,21,28	"	31	"	"
N90	1A2 to 1C4 "	7,22,23	21	"	3,4,5,26, 28	"	"	"	"
N91	1A3 to 1C4 "	7,23	3	"	4,5,21,22, 26,28	"	"	"	"
N92	1A4 to 1C4 "	28	7	"	3,4,5,21, 22,23,26	"	"	"	"
N93	1B1 to 1C4 "	3,4,21, 28	26	"	5,7,22,23,	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-0-20T 5:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N94	1B2 to 1C4	Propagation delay time (t_{PLH}), C_L = 150 pF	3,21,28	22	6	4,5,7,23, 26	5	31	ns	13,14
N95	1B3 to 1C4	"	3,28	23	"	4,5,7,21, 22,26	"	"	"	"
N96	1B4 to 1C4	"	7	28	"	3,4,5,21, 22,23,26	"	"	"	"
N97	2A1 to 2C4	"	8,13,30, 31	33	14	12,29,34, 36	"	"	"	"
N98	2A2 to 2C4	"	8,30,31	29	"	12,13,33, 34,36	"	"	"	"
N99	2A3 to 2C4	"	8,31	12	"	13,29,30, 33,34,36	"	"	"	"
N100	2A4 to 2C4	"	36	8	"	12,13,29, 30,31,33, 34	"	"	"	"
N101	2B1 to 2C4	"	12,29,33, 36	13	"	8,30,31,34	"	"	"	"
N102	2B2 to 2C4	"	12,29,36	30	"	8,13,31, 33,34	"	"	"	"
N103	2B3 to 2C4	"	12,36	31	"	8,13,29, 30,33,34	"	"	"	"
N104	2B4 to 2C4	"	8	36	"	12,13,29, 30,31,33, 34	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N105	1C1 to 1C4	Propagation delay time (t_{PLH}), C_L = 150 pF	3,4,7,21	5	6	22,23,26, 28	5	31	ns	13,14
N106	2C1 to 2C4	Propagation delay time (t_{PLH}), C_L = 150 pF	8,12,29, 33	34	14	13,30,31, 36	"	31	"	"
P1	1C1 to 1S1	Propagation delay time (t_{PHL}), C_L = 50 pF	3,4,7,21	5	25	22,23,26, 28	"	28	"	"
P2	1C1 to 1S2	"	"	"	2	"	"	"	"	"
P3	1C1 to 1S3	"	"	"	24	"	"	"	"	"
P4	1C1 to 1S4	"	"	"	27	"	"	"	"	"
P5	2C1 to 2S1	"	8,12,29, 33	34	32	13,30,31, 36	"	"	"	"
P6	2C1 to 2S2	"	"	"	9	"	"	"	"	"
P7	2C1 to 2S3	"	"	"	15	"	"	"	"	"
P8	2C1 to 2S4	"	"	"	35	"	"	"	"	"
P9	3C1 to 3S1	"	17,18	37	16	38,39	"	"	"	"
P10	3C1 to 3S2	"	17,18	37	19	38,39	"	"	"	"
P11	1A1 to 1S1	"	Open	4	25	3,5,7,21, 22,23,26, 28	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20261[illegible]-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P12	1A2 to 1S2 Propagation delay time (t_{PHL}), C_L = 50 pF	Open	21	2	3,4,5,7, 22,23,26, 28	5	28	ns	13,14
P13	1A3 to 1S3 "	"	3	24	4,5,7,21, 22,23,26, 28	"	"	"	"
P14	1A4 to 1S4 "	"	7	27	3,4,5,21, 22,23,26, 28	"	"	"	"
P15	1B1 to 1S1 "	"	26	25	3,4,5,7, 21,22,23, 28	"	"	"	"
P16	1B2 to 1S2 "	"	22	2	3,4,5,7, 21,23,26, 28	"	"	"	"
P17	1B3 to 1S3 "	"	23	24	3,4,5,7, 21,22,26, 28	"	"	"	"
P18	1B4 to 1S4 "	"	28	27	3,4,5,7, 21,22,23, 26	"	"	"	"
P19	2A1 to 2S1 "	"	33	32	8,12,13, 29,30,31, 34,36	"	"	"	"
P20	2A2 to 2S2 "	"	29	9	8,12,13, 30,31,33, 34,36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P21	2A3 to 2S3 Propagation delay time (t_{PHL}), C_L = 50 pF	Open	12	15	8,13,29, 30,31,33, 34,36	5	28	ns	13,14
P22	2A4 to 2S4 "	"	8	35	12,13,29, 30,31,33, 34,36	"	"	"	"
P23	2B1 to 2S1 "	"	13	32	8,12,29, 30,31,33, 34,36	"	"	"	"
P24	2B2 to 2S2 "	"	30	9	8,12,13, 29,31,33, 34,36	"	"	"	"
P25	2B3 to 2S3 "	"	31	15	8,12,13, 29,30,33, 34,36	"	"	"	"
P26	2B4 to 2S4 "	"	36	35	8,12,13, 29,30,31, 33,34	"	"	"	"
P27	3A1 to 3S1 "	"	17	16	18,37,38, 39	"	"	"	"
P28	3A2 to 3S2 "	"	18	19	17,37,38, 39	"	"	"	"
P29	3B1 to 3S1 "	"	38	16	17,18,37, 39	"	"	"	"
P30	3B2 to 3S2 "	"	39	19	17,18,37, 38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P31	3A1 to 3C2	Propagation delay time (t_{PHL}), C_L = 50 pF	18,38	17	40	37,39	5	28	ns	13,14
P32	3A2 to 3C2	"	39	18	"	17,37,38	"	"	"	"
P33	3B1 to 3C2	"	17,39	38	"	18,37	"	"	"	"
P34	3B2 to 3C2	"	18	39	"	17,37,38	"	"	"	"
P35	3C1 to 3C2	"	17,18	37	"	38,39	"	"	"	"
P36	1A1 to 1C4	"	7,22,23, 26	4	6	3,5,21,28	"	21	"	"
P37	1A2 to 1C4	"	7,22,23	21	"	3,4,5,26, 28	"	"	"	"
P38	1A3 to 1C4	"	7,23	3	"	4,5,21,22, 26,28	"	"	"	"
P39	1A4 to 1C4	"	28	7	"	3,4,5,21, 22,23,26	"	"	"	"
P40	1B1 to 1C4	"	3,4,21, 28	26	"	5,7,22,23	"	"	"	"
P41	1B2 to 1C4	"	3,21,28	22	"	4,5,7,23, 26	"	"	"	"
P42	1B3 to 1C4	"	3,28	23	"	4,5,7,21, 22,26	"	"	"	"
P43	1B4 to 1C4	"	7	28	"	3,4,5,21, 22,23,26	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P44	2A1 to 2C4	Propagation delay time (t_{PHL}), C_L = 50 pF	8,13,30, 31	33	14	12,29,34, 36	5	21	ns	13,14
P45	2A2 to 2C4	"	8,30,31	29	"	12,13,33, 34,36	"	"	"	"
P46	2A3 to 2C4	"	8,31	12	"	13,29,30, 33,34,36	"	"	"	"
P47	2A4 to 2C4	"	36	8	"	12,13,29, 30,31,33, 34	"	"	"	"
P48	2B1 to 2C4	"	12,29,33, 36	13	"	8,30,31,34	"	"	"	"
P49	2B2 to 2C4	"	12,29,36	30	"	8,13,31, 33,34	"	"	"	"
P50	2B3 to 2C4	"	12,36	31	"	8,13,29, 30,33,34	"	"	"	"
P51	2B4 to 2C4	"	8	36	"	12,13,29, 30,31,33, 34	"	"	"	"
P52	1C1 to 1C4	"	3,4,7,21	5	6	22,23,26, 28	"	"	"	"
P53	2C1 to 2C4	"	8,12,29, 33	34	14	13,30,31, 36	"	"	"	"
P54	1C1 to 1S1	Propagation delay time (t_{PHL}), C_L = 150 pF	3,4,7,21	5	25	22,23,26, 28	"	38	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-0-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
P55	1C1 to 1S2	Propagation delay time (t_{PHL}), C_L = 150 pF	3,4,7,21	5	2	22,23,26, 28	5	38	ns	13,14
P56	1C1 to 1S3	"	"	"	24	"	"	"	"	"
P57	1C1 to 1S4	"	"	"	27	"	"	"	"	"
P58	2C1 to 2S1	"	8,12,29, 33	34	32	13,30,31, 36	"	"	"	"
P59	2C1 to 2S2	"	"	"	9	"	"	"	"	"
P60	2C1 to 2S3	"	"	"	15	"	"	"	"	"
P61	2C1 to 2S4	"	"	"	35	"	"	"	"	"
P62	3C1 to 3S1	"	17,18	37	16	38,39	"	"	"	"
P63	3C1 to 3S2	"	17,18	37	19	38,39	"	"	"	"
P64	1A1 to 1S1	"	Open	4	25	3,5,7,21, 22,23,26, 28	"	"	"	"
P65	1A2 to 1S2	"	"	21	2	3,4,5,7, 22,23,26, 28	"	"	"	"
P66	1A3 to 1S3	"	"	3	24	4,5,7,21, 22,23,26, 28	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P67	1A4 to 1S4 Propagation delay time (t_{PHL}), C_L = 150 pF	Open	7	27	3,4,5,21, 22,23,26, 28	5	38	ns	13,14
P68	1B1 to 1S1 "	"	26	25	3,4,5,7, 21,22,23, 28	"	"	"	"
P69	1B2 to 1S2 "	"	22	2	3,4,5,7, 21,23,26, 28	"	"	"	"
P70	1B3 to 1S3 "	"	23	24	3,4,5,7, 21,22,26, 28	"	"	"	"
P71	1B4 to 1S4 "	"	28	27	3,4,5,7, 21,22,23, 26	"	"	"	"
P72	2A1 to 2S1 "	"	33	32	8,12,13, 29,30,31, 34,36	"	"	"	"
P73	2A2 to 2S2 "	"	29	9	8,12,13, 30,31,33, 34,36	"	"	"	"
P74	2A3 to 2S3 "	"	12	15	8,13,29, 30,31,33, 34,36	"	"	"	"
P75	2A4 to 2S4 "	"	8	35	12,13,29, 30,31,33, 34,36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P76	2B1 to 2S1 Propagation delay time (t_{PHL}), C_L = 150 pF	Open	13	32	8,12,29, 30,31,33, 34,36	5	38	ns	13,14
P77	2B2 to 2S2 "	"	30	9	8,12,13, 29,31,33, 34,36	"	"	"	"
P78	2B3 to 2S3 "	"	31	15	8,12,13, 29,30,33, 34,36	"	"	"	"
P79	2B4 to 2S4 "	"	36	35	8,12,13, 29,30,31, 33,34	"	"	"	"
P80	3A1 to 3S1 "	"	17	16	18,37,38, 39	"	"	"	"
P81	3A2 to 3S2 "	"	18	19	17,37,38, 39	"	"	"	"
P82	3B1 to 3S1 "	"	38	16	17,18,37, 39	"	"	"	"
P83	3B2 to 3S2 "	"	39	19	17,18,37, 38	"	"	"	"
P84	3A1 to 3C2 "	18,38	17	40	37,39	"	"	"	"
P85	3A2 to 3C2 "	39	18	"	17,37,38	"	"	"	"
P86	3B1 to 3C2 "	17,39	38	"	18,37	"	"	"	"
P87	3B2 to 3C2 "	18	39	"	17,37,38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P88	3C1 to 3C2 Propagation delay time (t_{PHL}), C_L = 150 pF	17,18	37	40	38,39	5	38	ns	13,14
P89	1A1 to 1C4 "	7,22,23, 26	4	6	3,5,21,28	"	31	"	"
P90	1A2 to 1C4 "	7,22,23	21	"	3,4,5,26, 28	"	"	"	"
P91	1A3 to 1C4 "	7,23	3	"	4,5,21,22, 26,28	"	"	"	"
P92	1A4 to 1C4 "	28	7	"	3,4,5,21, 22,23,26	"	"	"	"
P93	1B1 to 1C4 "	3,4,21, 28	26	"	5,7,22,23	"	"	"	"
P94	1B2 to 1C4 "	3,21,28	22	"	4,5,7,23, 26	"	"	"	"
P95	1B3 to 1C4 "	3,28	23	"	4,5,7,21, 22,26	"	"	"	"
P96	1B4 to 1C4 "	7	28	"	3,4,5,21, 22,23,26	"	"	"	"
P97	2A1 to 2C4 "	8,13,30, 31	33	14	12,29,34, 36	"	"	"	"
P98	2A2 to 2C4 "	8,30,31	29	"	12,13,33, 34,36	"	"	"	"
P99	2A3 to 2C4 "	8,31	12	"	13,29,30, 33,34,36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20261 0-20T15:38Z
Check the source to verify that this is the current version before use.

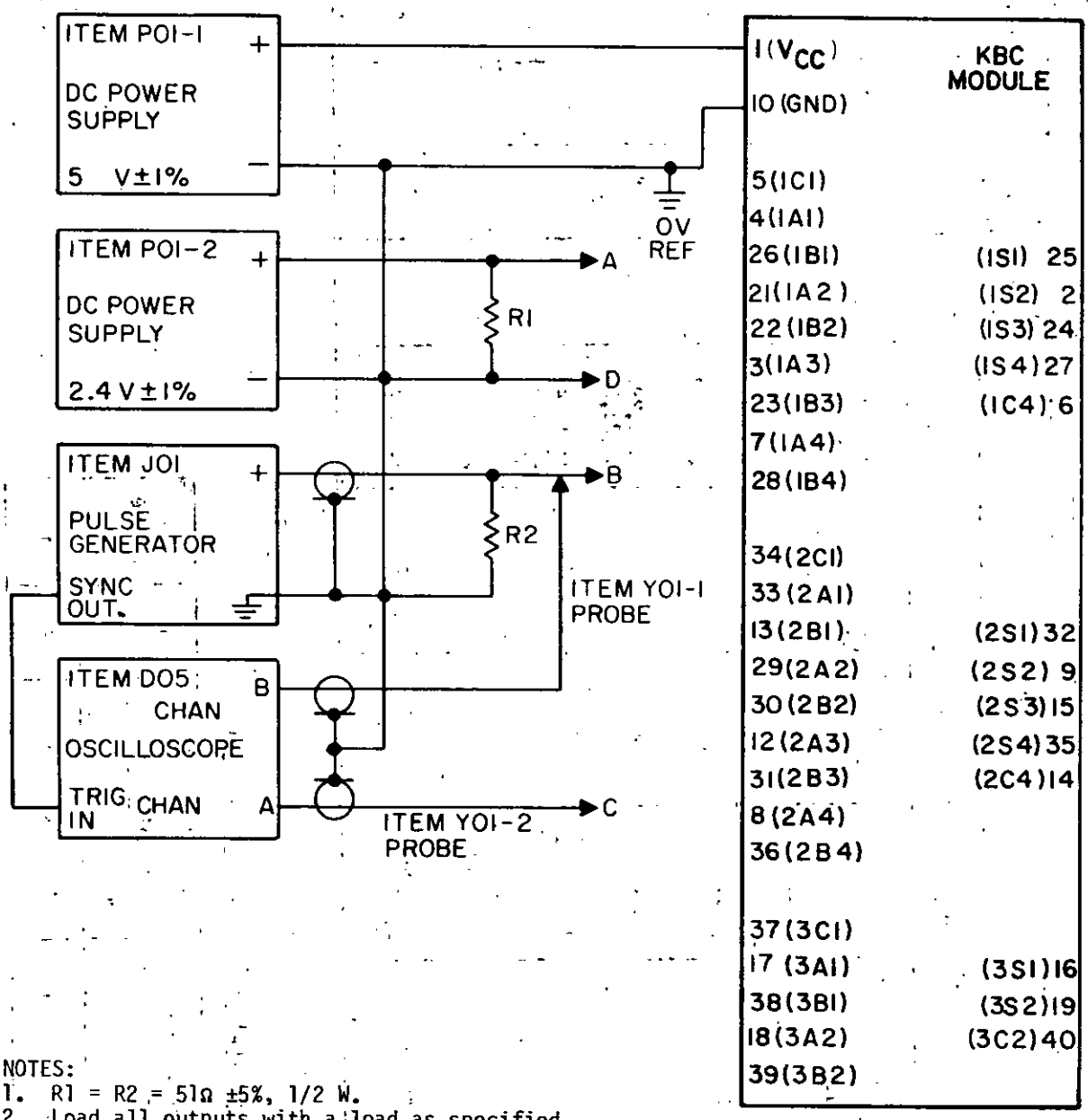
TABLE X. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P100	2A4 to 2C4 Propagation delay time (t_{PHL}), C_L = 150 pF	36	8	14	12,13,29, 30,31,33, 34	5	31	ns	13,14
P101	2B1 to 2C4 "	12,29,33, 36	13	"	8,30,31,34	"	"	"	"
P102	2B2 to 2C4 "	12,29,36	30	"	8,13,31, 33,34	"	"	"	"
P103	2B3 to 2C4 "	12,36	31	"	8,13,29, 30,33,34	"	"	"	"
P104	2B4 to 2C4 "	8	36	"	12,13,29, 30,31,33, 34	"	"	"	"
P105	1C1 to 1C4 "	3,4,7,21	5	6	22,23,26, 28	"	"	"	"
P106	2C1 to 2C4 "	8,12,29, 33	34	14	13,30,31, 36	"	"	"	"

TABLE XI. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
Q	Power supply filter capacitance	1.6	3.0	µF	15

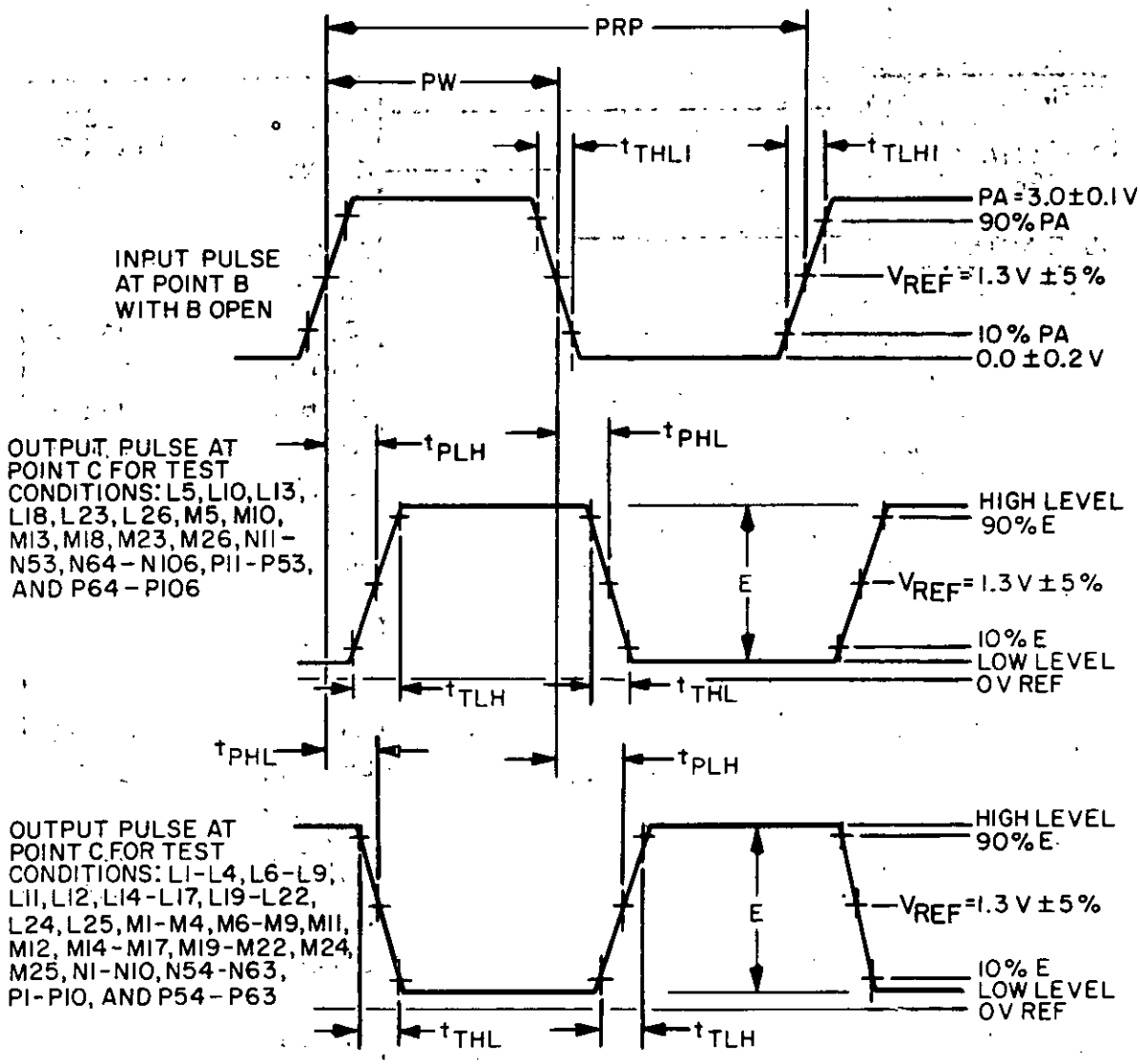
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a load as specified on figure 10.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table X.

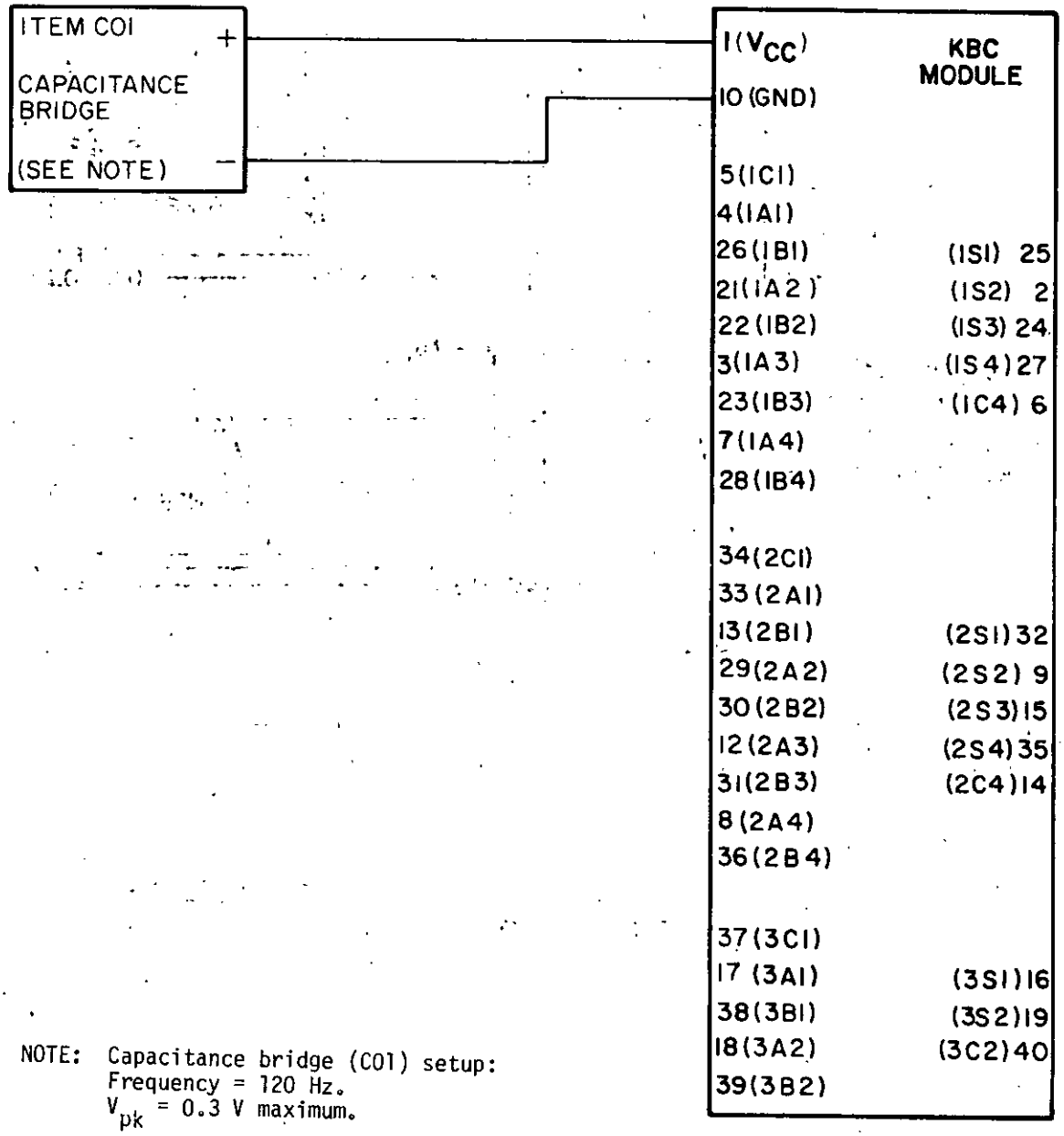
Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 2$ ns.
2. All voltage levels are referenced to ground.
3. Before taking measurement, verify that the output high level is $\geq$ 2.4 V and the output low level is $\leq$ 0.4 V.

* FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: Capacitance bridge (C01) setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

*FIGURE 15. Power supply filter capacitance test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XII. Output crosstalk (high-level and low-level) test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1S1	Output crosstalk, high level	E	Open	F,G	25	0	400	mV	16,17
R2	1S2	"	"	"	"	2	"	"	"	"
R3	1S3	"	"	"	"	24	"	"	"	"
R4	1S4	"	"	"	"	27	"	"	"	"
R5	1C4	"	"	"	"	6	"	"	"	"
R6	2S1	"	F	"	E,G	32	"	"	"	"
R7	2S2	"	"	"	"	9	"	"	"	"
R8	2S3	"	"	"	"	15	"	"	"	"
R9	2S4	"	"	"	"	35	"	"	"	"
R10	2C4	"	"	"	"	14	"	"	"	"
R11	3S1	"	G	"	E,F	16	"	"	"	"
R12	3S2	"	"	"	"	19	"	"	"	"
R13	3C2	"	"	"	"	40	"	"	"	"
S1	1S1	Output crosstalk, low level	Open	E	F,G	25	"	"	"	"
S2	1S2	"	"	"	"	2	"	"	"	"
S3	1S3	"	"	"	"	24	"	"	"	"
S4	1S4	"	"	"	"	27	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

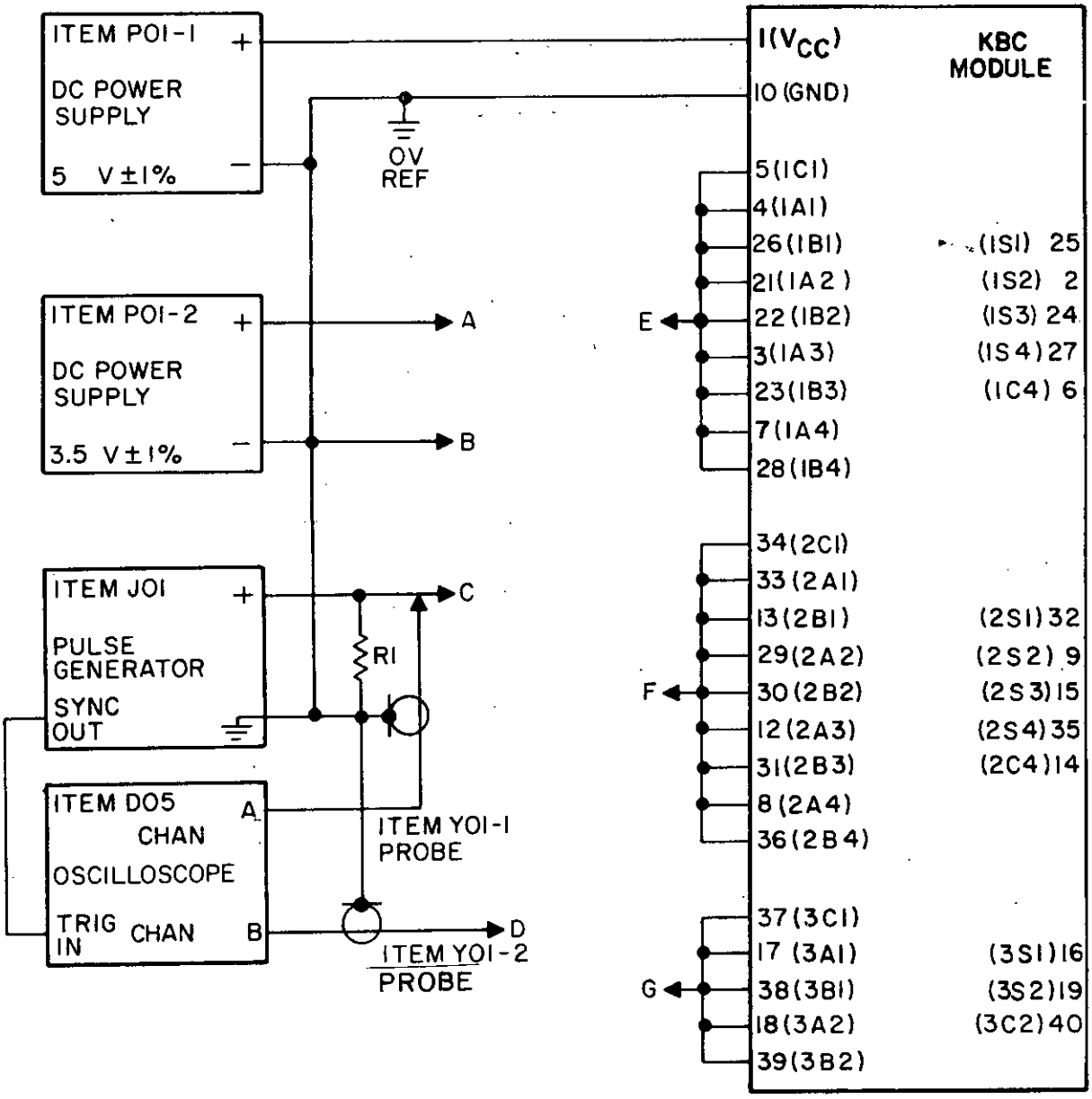
TABLE XII. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
S5	1C4	Output crosstalk, low level	Open	E	F,G	6	0	400	mV	16,17
S6	2S1	"	"	F	E,G	32	"	"	"	"
S7	2S2	"	"	"	"	9	"	"	"	"
S8	2S3	"	"	"	"	15	"	"	"	"
S9	2S4	"	"	"	"	35	"	"	"	"
S10	2C4	"	"	"	"	14	"	"	"	"
S11	3S1	"	"	G	E,F	16	"	"	"	"
S12	3S2	"	"	"	"	19	"	"	"	"
S13	3C2	"	"	"	"	40	"	"	"	"

TABLE XIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	400	mV	18,19

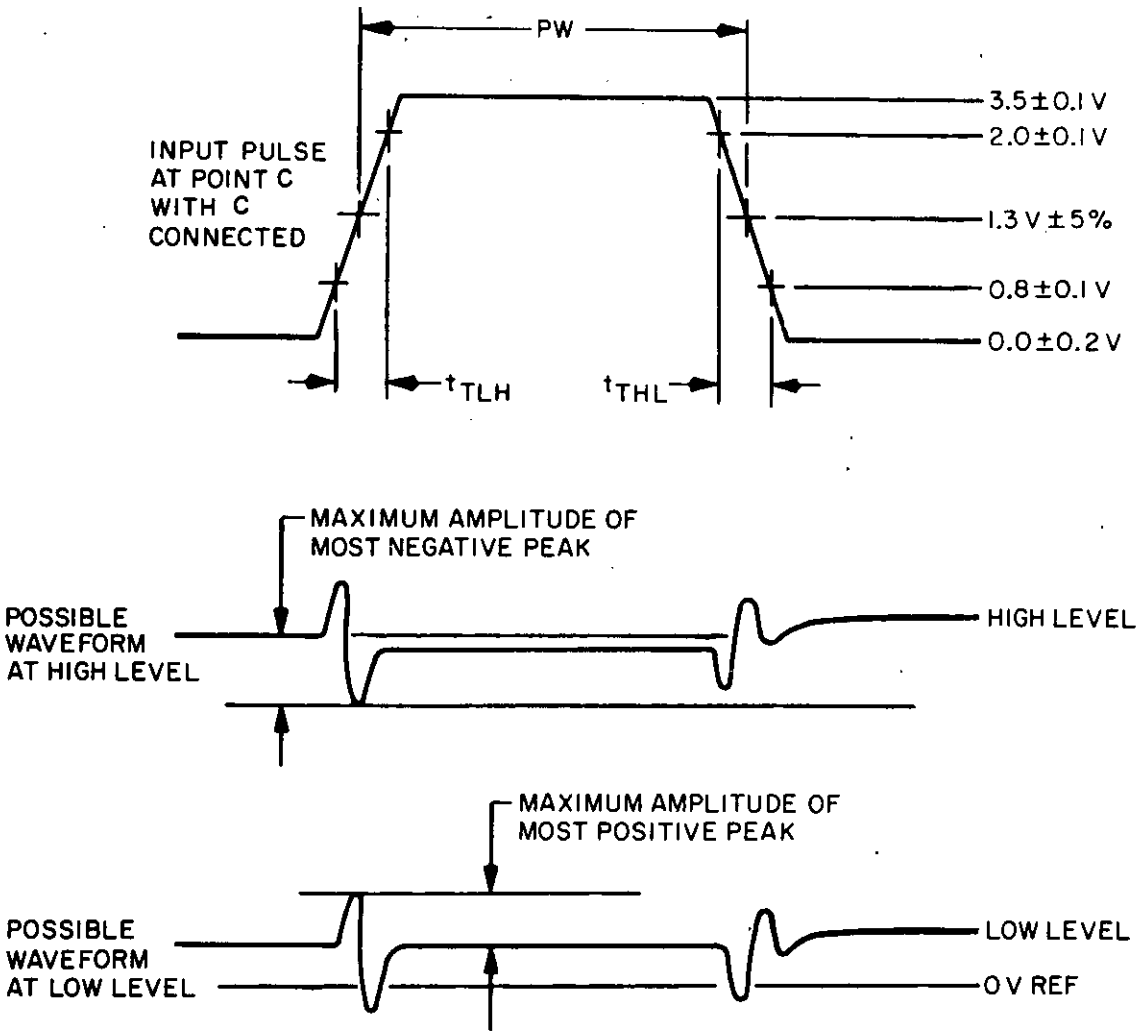
Source: https://assist.dla.mil -- Downloaded: 20261 0-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a load as specified on figure 10.

FIGURE 16. Output crosstalk (high-level and low-level) test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:

PW = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PRR = 1 MHz ±10%.

FIGURE 17. Output crosstalk (high-level and low-level) test circuit waveforms for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

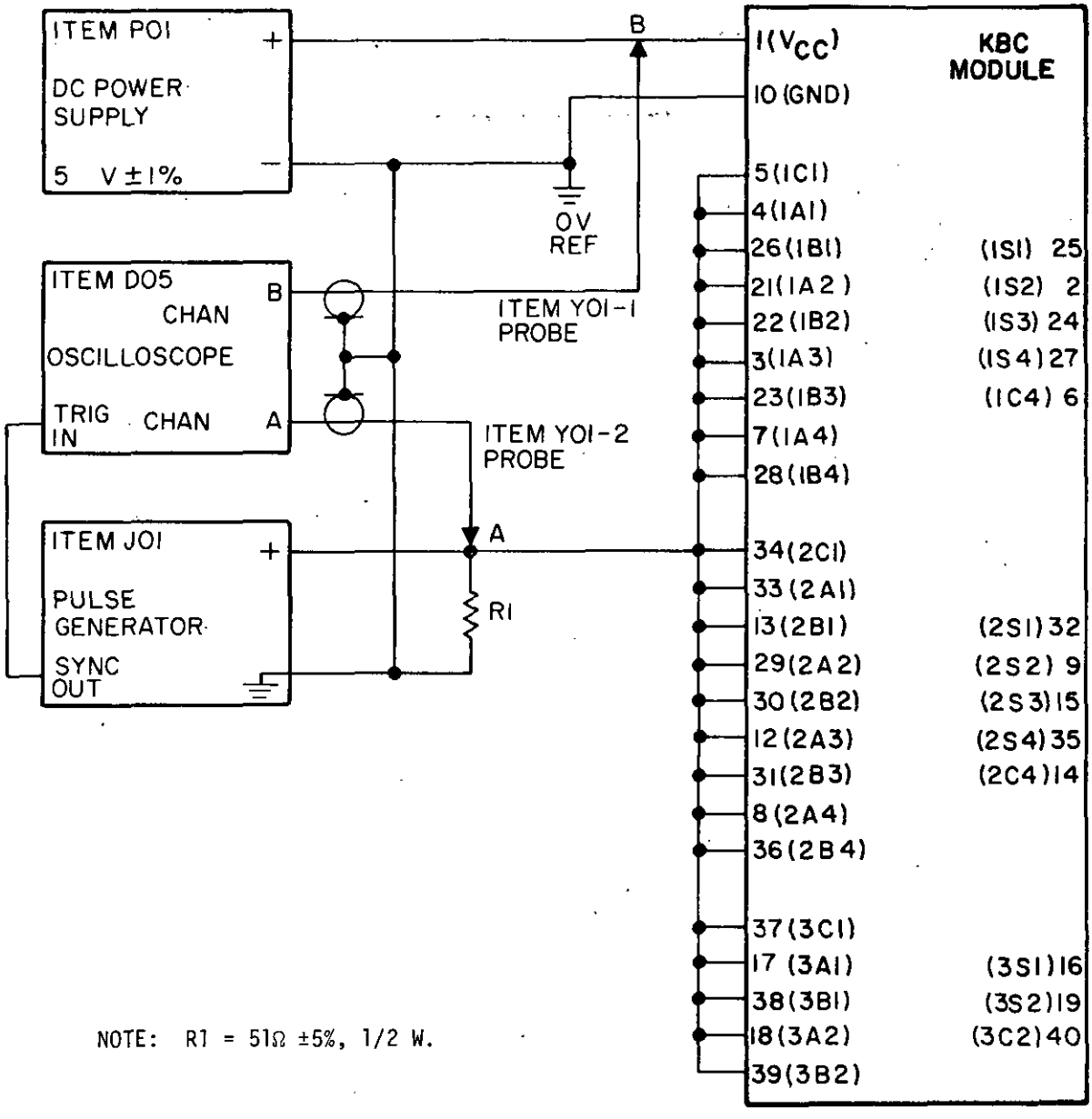
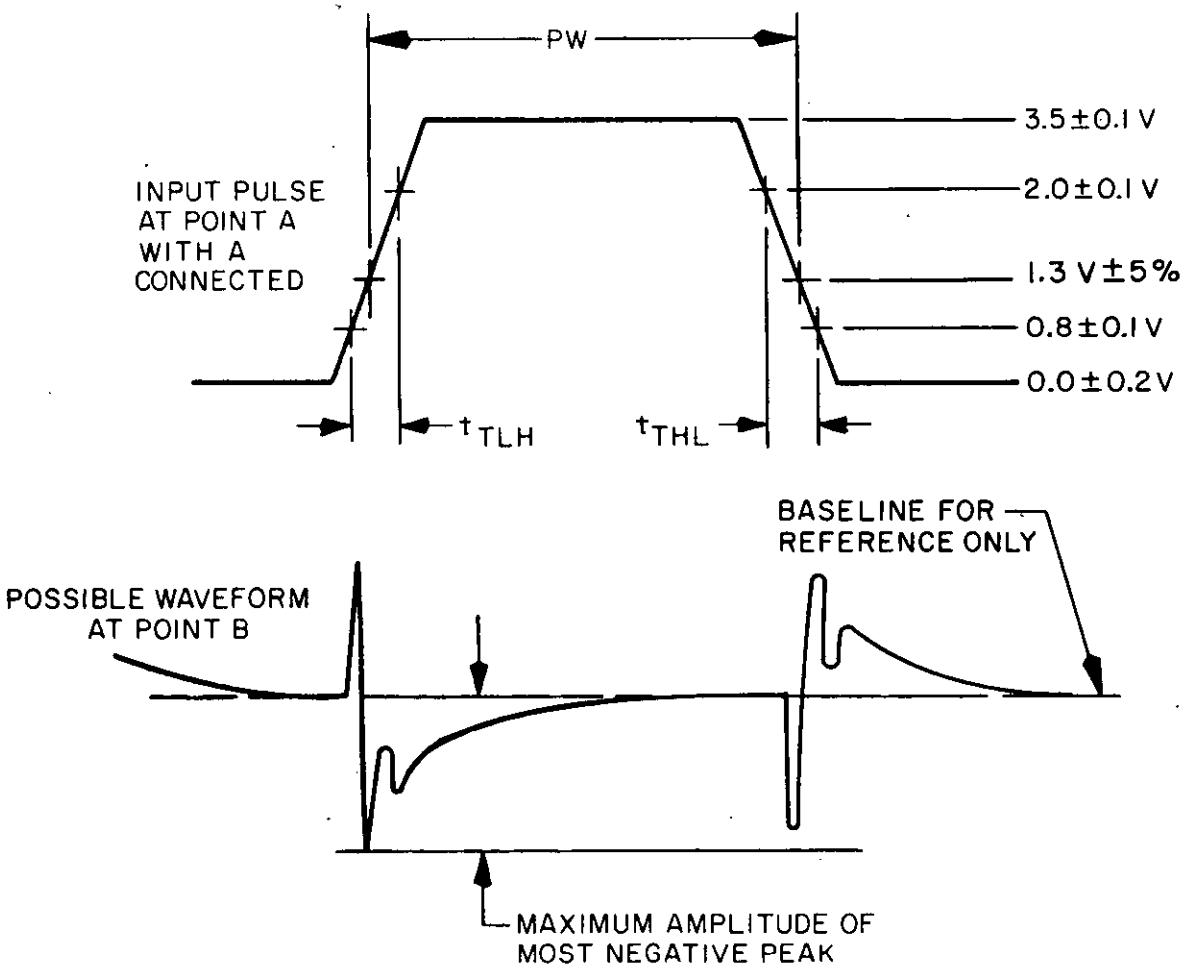


FIGURE 18. Power supply transient amplitude test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:

PW = 500 ns ±10%.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
PRR = 1.0 MHz ±10%.

FIGURE 19. Power supply transient amplitude test circuit waveforms for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XIV. Input capacitance test.

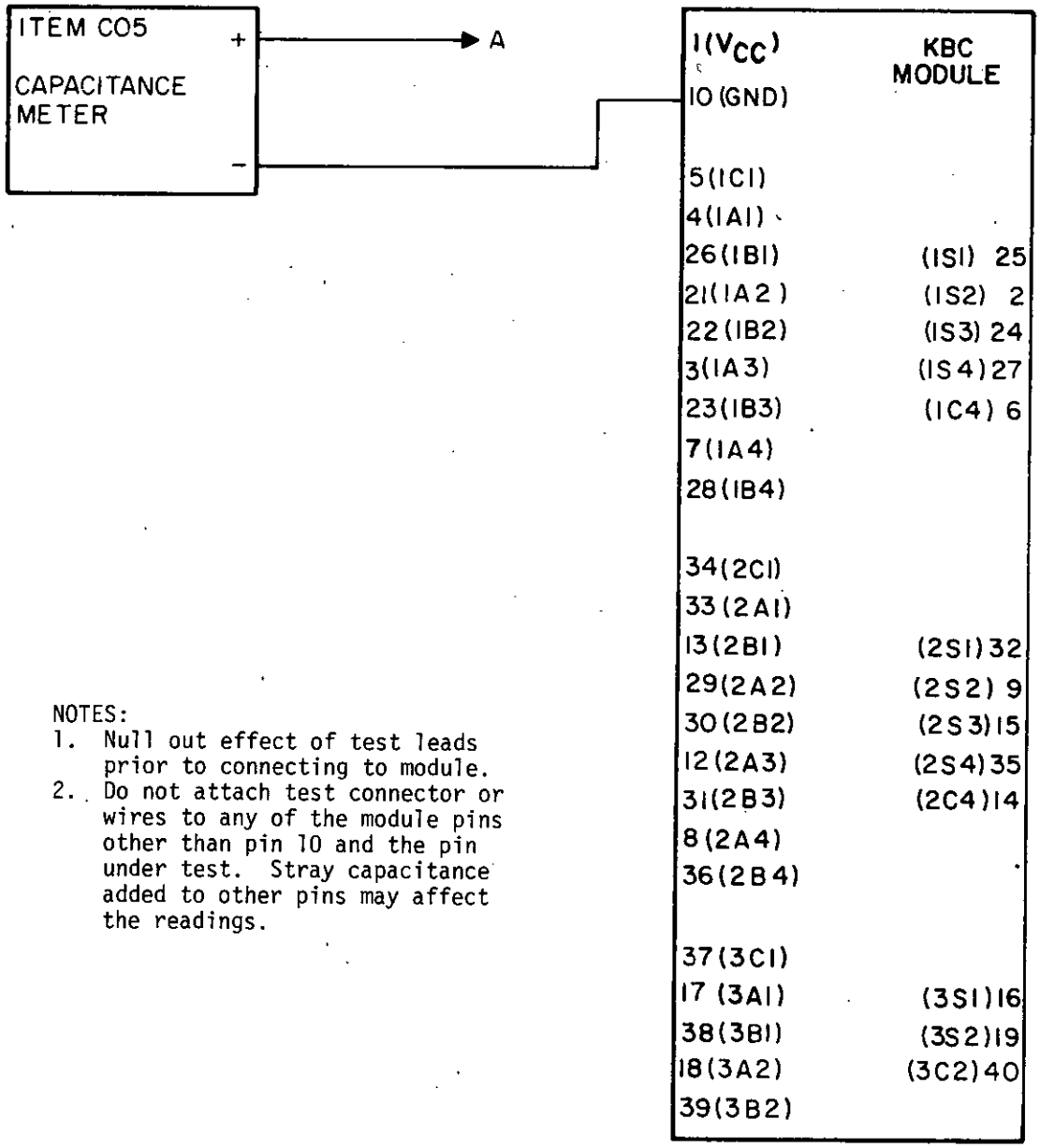
Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
U1	1C1	Input capacitance	5	2	15	pF	20
U2	1A1	"	4	"	"	"	"
U3	1B1	"	26	"	"	"	"
U4	1A2	"	21	"	"	"	"
U5	1B2	"	22	"	"	"	"
U6	1A3	"	3	"	"	"	"
U7	1B3	"	23	"	"	"	"
U8	1A4	"	7	"	"	"	"
U9	1B4	"	28	"	"	"	"
U10	2C1	"	34	"	"	"	"
U11	2A1	"	33	"	"	"	"
U12	2B1	"	13	"	"	"	"
U13	2A2	"	29	"	"	"	"
U14	2B2	"	30	"	"	"	"
U15	2A3	"	12	"	"	"	"
U16	2B3	"	31	"	"	"	"
U17	2A4	"	8	"	"	"	"
U18	2B4	"	36	"	"	"	"
U19	3C1	"	37	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

TABLE XIV. Input capacitance test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
U20	3A1	Input capacitance	17	2	15	pF	20
U21	3B1	"	38	"	"	"	"
U22	3A2	"	18	"	"	"	"
U23	3B2	"	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20261 0-20T15:38Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out effect of test leads prior to connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

* FIGURE 20. Input capacitance test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

* TABLE XV. Functional test.

Test condition	Parameter	Connect			Output logic states													Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1S1	1S2	1S3	1S4	1C4	2S1	2S2	2S3	2S4	2C4	3S1	3S2	3C2	Min	Max		
V1	Functional	3,7,8,12,17, 23,31,38,39	4,5,13,18,21, 22,26,28,29, 30,33,34,36, 37	1/ 2/	L	L	L	L	H	L	L	L	L	H	L	L	H	3/			21
V2	"	3,12,17,21, 23,29,31,37	4,5,7,8,13, 18,22,26,28, 30,33,34,36, 38,39	"	L	H	L	H	L	L	H	L	H	L	L	H	L	"			"
V3	"	3,4,12,18, 21,23,28, 29,31,33, 36,39	5,7,8,13,17, 22,26,30,34, 37,38	"	H	H	L	L	H	H	H	L	L	H	L	L	H	"			"
V4	"	4,5,13,17, 26,33,34, 37,38,39	3,7,8,12,18, 21,22,23,28, 29,30,31,36	"	H	H	L	L	L	H	H	L	L	L	H	L	H	"			"
V5	"	3,4,5,7,8, 12,13,23, 26,28,31, 33,34,36, 38,39	17,18,21,22, 29,30,37	"	H	H	L	H	H	H	H	L	H	H	H	H	L	"			"
V6	"	3,4,5,7,8, 12,33,34, 37,39	13,17,18,21, 22,23,26,28, 29,30,31,36, 38	"	L	H	H	H	L	L	H	H	H	L	H	H	L	"			"
V7	"	3,4,5,7,8, 12,13,17, 21,22,23, 26,29,30, 31,33,34, 39	18,28,36,37, 38	"	H	H	H	L	H	H	H	H	L	H	H	H	L	"			"

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

* TABLE XV. Functional test - Continued.

Test condition	Parameter	Connect			Output logic states													Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1S1	1S2	1S3	1S4	1C4	2S1	2S2	2S3	2S4	2C4	3S1	3S2	3C2	Min	Max		
V8	Functional	3,4,5,7,8, 12,13,21,22, 26,29,30,33, 34	17,18,23,28 31,36,37,38, 39	1/ 2/	H	H	L	L	H	H	H	L	L	H	L	L	L	3/			21
V9	"	4,5,7,8,13, 17,18,21,22, 26,29,30,33, 34,37	3,12,23,28, 31,36,38,39	"	H	H	H	H	L	H	H	H	H	L	L	L	H	"			"
V10	"	4,5,7,8,13, 17,18,26,33, 34,37,39	3,12,21,22, 23,28,29,30, 31,36,38	"	H	H	L	H	L	H	H	L	H	L	L	H	H	"			"
V11	"	5,22,23,28, 30,31,34,36	3,4,7,8,12, 13,21,26,29, 33	"	H	H	H	H	L	H	H	H	H	L				"			"
V12	"	5,13,22,23, 26,28,30,31, 34,36	3,4,7,8,12, 21,29,33	"	L	L	L	L	H	L	L	L	L	H				"			"
V13	"	5,21,22,23, 28,29,30,31, 34,36	3,4,7,8,12, 13,26,33	"	H	L	L	L	H	H	L	L	L	H				"			"
V14	"	13,22,23,26, 28,30,31,36	3,4,5,7,8, 12,21,29,33, 34	"	H	H	H	H	L	H	H	H	H	L				"			"
V15	"	7,8,28,36	3,4,5,12,13, 21,22,23,26, 29,30,31,33, 34	"	L	L	L	L	H	L	L	L	L	H				"			"

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

* TABLE XV. Functional test - Continued.

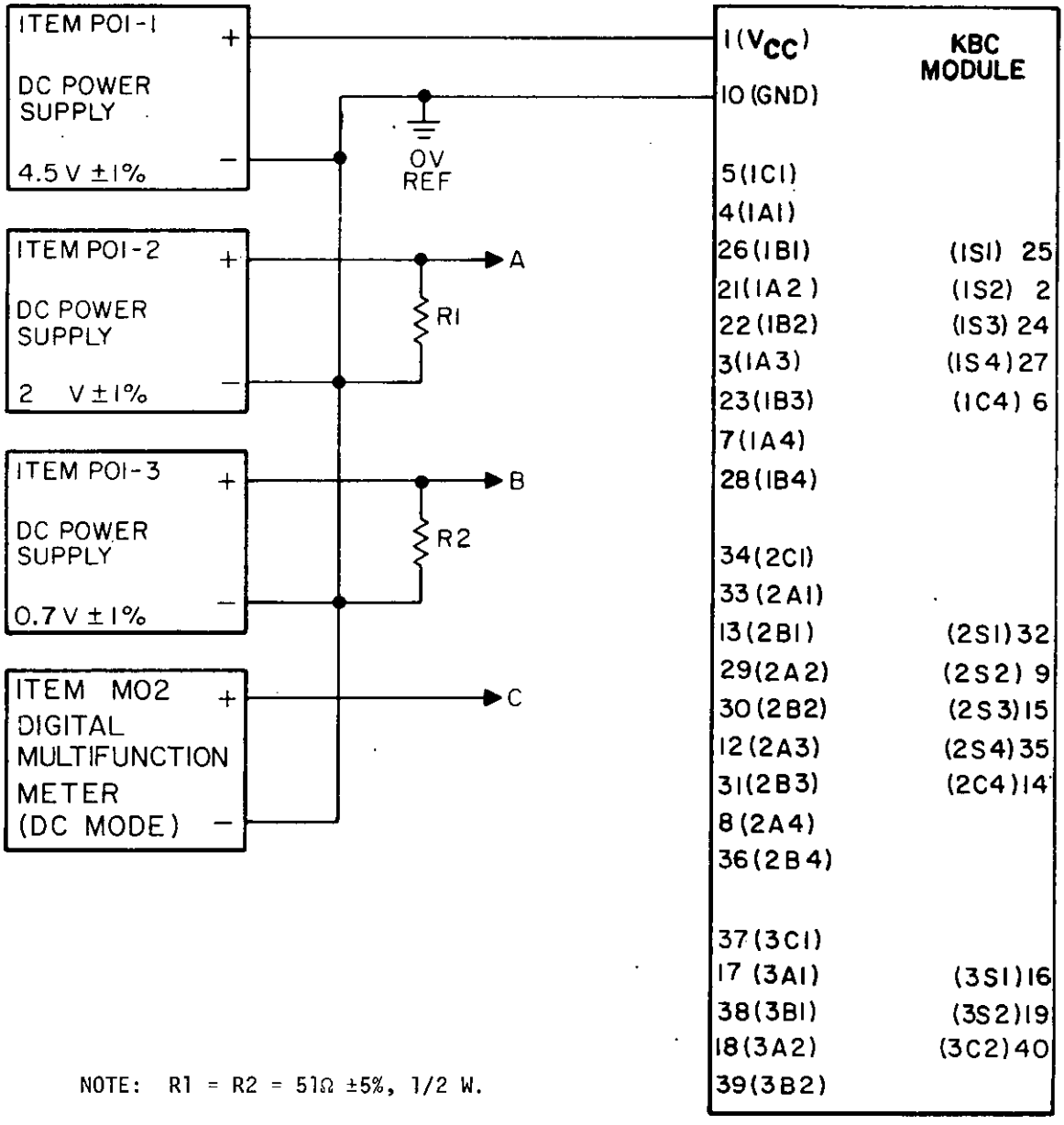
Test condition	Parameter	Connect			Output logic states													Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	1S1	1S2	1S3	1S4	1C4	2S1	2S2	2S3	2S4	2C4	3S1	3S2	3C2	Min	Max		
V16	Functional	7,8,13,22, 23,26,28,30, 31,36	3,4,5,12,21, 29,33,34	1/ 2/	H	H	H	L	H	H	H	H	L	H				3/			21
V17	"	4,13,22,23, 26,28,30,31, 33,36	3,5,7,8,12, 21,29,34	" "	L	L	L	L	H	L	L	L	L	H				"			"
V18	"	13,21,22,23, 26,28,29,30, 31,36	3,4,5,7,8, 12,33,34	" "	H	L	L	L	H	H	L	L	L	H				"			"

1/ For test conditions V1-V18, verification of the output logic states in footnote 3 is obtained by connecting item M02, point C, to each output pin in succession after input conditions have been accomplished.

2/ 1S1, 1S2, 1S3, 1S4, 1C4, 2S1, 2S2, 2S3, 2S4, 2C4, 3S1, 3S2, and 3C2 output logic states are measured at output pins 25, 2, 24, 27, 6, 32, 9, 15, 35, 14, 16, 19, and 40 respectively.

3/ Output logic states:
L = Low level = 0.0 to 0.4 V.
H = High level = 2.4 to 4.5 V.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:38Z
Check the source to verify that this is the current version before use.



NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

* FIGURE 21. Functional test circuit for table XV.

☆ U.S. GOVERNMENT PRINTING OFFICE: 1983—605-034/666

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-28787/39B(EC), MODULES, STANDARD ELECTRONIC ADDER, DIGITAL, KEY CODE KRC

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:38Z
Check the source to verify that this is the current version before use.